BaharequeModular

Open-Source Prefabricated Panel System

Designed by Jan Reuteler
janreuteler@icloud.com

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BahareqeModular — Design Philosophy

Origin — Learning by Doing

This system was not designed at a desk. It grew out of hands-on experience with permaculture and bioconstruction courses in rural Colombia — building with guadua, mixing mortar, plastering walls, and seeing firsthand what works, what fails, and why.

In those courses, you learn to appreciate the extraordinary qualities of natural materials: the strength of guadua bamboo, the breathability of lime, the thermal comfort of earthen walls, the beauty of hand-finished surfaces. You also learn the limits. Bioconstruction projects — however well-intentioned — often struggle with:

- **Durability.** Earth walls erode in heavy tropical rain. Untreated bamboo is eaten by insects within years. Buildings that took months of community effort start degrading within a decade.
- **Structural safety.** Traditional bahareque and adobe perform poorly in earthquakes unless carefully engineered. Most community-built projects don't have access to structural engineers. The 1999 Armenia earthquake in Colombia killed over 1,000 people — many in unreinforced masonry and poorly built traditional structures.
- **Building permits and insurance.** In most countries, earth and bamboo buildings cannot be formally permitted or insured. Families who build with natural materials often have no legal protection for their home.
- **Finish quality.** Bioconstruction workshops produce beautiful demonstration walls. But scaling to a full house — with consistent surfaces, integrated electrical, plumbing, and a finish that withstands years of weather — is a different challenge. Many projects look rough compared to conventional construction, reinforcing the perception that "natural = poor quality."
- **Reproducibility.** Each bioconstruction project depends on the skill of the builders present. There is no standardized panel, no production line, no quality control process. When the experienced builder leaves, the quality drops.

These are not failures of the materials or the philosophy — they are failures of the system around them. Guadua is as strong as steel in tension. Lime plaster has protected buildings for millennia. The problem is not the ingredients. The problem is that no one had packaged them into a **repeatable, quality-controlled, structurally engineered, code-compliant system** that a community can produce without depending on a master builder.

That is what this project attempts to solve.

The Problem

Most "affordable housing" in the developing world faces a difficult trade-off:

- **Traditional construction** (bamboo, adobe, wattle-and-daub) is beautiful, breathable, culturally meaningful, and uses local materials — but it cracks in earthquakes, degrades in heavy rain, and rarely gets building permits or insurance.
- **Modern construction** (concrete block, reinforced concrete frame) gets permits and survives earthquakes — but it's expensive, requires specialized labor, looks identical everywhere, and has no connection to place or culture.

The result: billions live in buildings that either fail structurally or fail the people who inhabit them.

A Worldwide Tradition

This system stands on the shoulders of millennia of building knowledge. Across every continent, cultures developed frame-and-infill construction — a structural frame filled with natural materials. These techniques evolved independently because they work. They are earthquake-resistant, thermally comfortable, repairable, and beautiful.

Bahareque (Colombia, Ecuador, Central America)

Bamboo frame (guadua) filled with mud, mortar, or wattle. The traditional building method of Colombia's Eje Cafetero region, recognized by UNESCO as cultural heritage. Bahareque buildings survived the 1999 Armenia earthquake far better than unreinforced masonry. The Coffee Cultural Landscape of Colombia — a UNESCO World Heritage site — is defined by its bahareque architecture. This panel system is a direct evolution of bahareque: same materials, same philosophy, engineered for modern performance.

Quincha (Peru, Chile, Argentina)

Bamboo or cane frame with mud plaster, used since pre-Columbian times. Lima's colonial center was rebuilt in quincha after the 1746 earthquake — the flexible frames survived subsequent earthquakes that destroyed rigid masonry buildings. Quincha is still used in rural Peru and is experiencing a revival in sustainable architecture circles.

Wattle-and-Daub (Global — Europe, Africa, Asia, Americas)

The oldest known building technique, dating back 6,000+ years. Woven sticks (wattle) coated with mud or clay (daub). Found in medieval English cottages, West African compounds, Japanese *tsuchikabe* walls, and Mesoamerican structures. Many wattle-and-daub buildings in Europe have stood for 500+ years. The principle — a lattice of organic material encased in a mineral matrix — is exactly what this panel system uses.

Fachwerk / Colombage (Germany, France, Northern Europe)

Timber frame with infill panels (brick, straw, or mud). Known as *Fachwerk* in Germany, *colombage* in France, *half-timbering* in England. Thousands of buildings from the 14th–18th century still stand and are inhabited today. The key insight: the frame carries the structure, the infill provides enclosure. This is the same principle as the steel frame + bahareque panel.

Dhajji-Dewari (Kashmir, Pakistan, Northern India)

Timber frame with stone or mud infill, used in the Himalayan seismic zone. The word means "patchwork quilt wall" in Kashmiri. Dhajji-dewari buildings survived the devastating 2005 Kashmir earthquake while modern concrete structures collapsed. The closely spaced timber members and diagonal bracing create a ductile, energy-absorbing structure — the same principle behind the diagonal bamboo bracing in this panel system.

Gaiola Pombalina (Lisbon, Portugal)

After the catastrophic 1755 Lisbon earthquake and tsunami, the Marquis of Pombal commissioned a new building system: a timber cage (*gaiola*) with diagonal bracing inside masonry walls. This was the world's first engineered seismic construction standard. The diagonal bracing — converting shear forces to tension — is a direct ancestor of the diagonal bamboo strips in this panel system. Buildings with the Pombaline cage survived subsequent earthquakes for over 250 years.

Bajareque (Honduras, El Salvador, Guatemala)

Central American variant of bahareque, using local bamboo or cane with mud infill. Bajareque is experiencing renewed interest as communities seek earthquake-resistant, affordable alternatives to concrete block.

Taquezal (Nicaragua)

Similar to bahareque — timber or bamboo frame filled with mud/straw mixture. Traditional in Nicaraguan colonial architecture. León and Granada's historic centers feature taquezal construction.

Tabique (Mexico, Philippines)

In Mexico: bamboo or reed frame (*carrizo*) with mud plaster, common in Oaxaca, Guerrero, and Chiapas. In the Philippines: bamboo-framed walls (*tabique pampanga*) used in colonial-era buildings. Both demonstrate the universality of bamboo-frame construction across tropical regions.

Tsuchikabe (Japan)

Traditional Japanese earthen walls built on a bamboo lattice (*komai*). Multiple layers of earth plaster are applied over the woven bamboo — coarse base coat, intermediate coat, and fine finish coat. The same layered approach (structural lattice + mineral encasement + fine plaster finish) used in this panel system.

Pau-a-Pique (Brazil)

Brazilian wattle-and-daub using wooden stakes (*paus*) woven with bamboo or vines and plastered with earth. Common in rural Brazil since colonial times. Known for thermal comfort in Brazil's varied climates.

This panel system takes the core insight shared by all these traditions — a lattice of organic material encased in a mineral matrix, supported by a structural frame — and engineers it for modern performance: galvanized steel instead of raw timber, treated bamboo instead of untreated cane, pozzolanic mortar instead of raw mud, screw-clamping instead of rope binding. The wisdom is ancient. The execution is modern. The result lasts 80–100 years.

The Hybrid Approach

This panel system resolves the trade-off between traditional and modern construction:

Traditional where it's seen. The exterior surface is lime plaster over bamboo and mortar — the same materials used in bahareque, quincha, wattle-and-daub, and tsuchikabe for centuries. The texture is handmade. The surface breathes. The building belongs to its place.

Modern where it matters. Inside each panel is a hot-dip galvanized steel T-bar frame engineered for seismic loads. This frame is invisible once the panel is finished. It carries the structural reality while the traditional materials carry the human experience.

Permaculture Integration

The system embodies permaculture principles:

- **Use local resources:** Guadua angustifolia grows naturally across tropical Latin America at 0–2,000 m altitude. It reaches harvest size in 4–5 years. One hectare can supply enough guadua for dozens of panels annually.
- **Turn problems into solutions:** Invasive star grass (*Cynodon nlemfuensis*), a common agricultural pest, is dried, chopped, and used as fiber reinforcement in lime plaster. Removing the invasive species improves the land; using it in construction closes the loop.
- **Build soil, not waste:** Excavated volcanic ash soil (Andisol) from foundations becomes premium growing medium for raised beds and food forest terracing.
- **Design for succession:** An 80–100 year building outlasts the builder. The next generation inherits a home, not a rebuild project.

Bioconstruction, Engineered

Bioconstruction is not nostalgia. It is the recognition that natural materials — bamboo, lime, earth, stone — have physical properties that modern materials struggle to match:

- **Lime plaster** is antifungal, breathable, self-healing (carbonation fills micro-cracks), and beautiful. No paint needed.
- **Guadua bamboo** has a tensile strength comparable to mild steel, grows without irrigation, and sequesters carbon as it grows.
- **Mortar with pozzolanic admixture** (volcanic ash, rice husk ash, or metakaolin) reduces alkalinity over time, extending the life of embedded organic materials.

The steel frame doesn't replace these materials — it supports them. The frame handles what bamboo and mortar cannot: concentrated point loads, moment connections, and code-compliant seismic resistance.

Not Just Affordable — Beautiful

This is not a compromise system. The finished wall — hand-troweled lime plaster over dense mortar — produces a surface quality that competes with construction techniques costing 5–10× more.

- **Lime plaster** has been the finish of choice for palaces, cathedrals, and luxury villas for millennia. The same material covering a Tuscan farmhouse or a Colombian hacienda covers these panels. Each wall has the subtle texture of handwork — no two surfaces are identical.
- **Lime wash** develops a depth and warmth that paint cannot match. It interacts with light differently at every hour. It ages gracefully — gathering a patina rather than peeling.
- **The wall is solid.** Knock on it — it sounds and feels like stone. There is no hollow cavity, no drywall flex. The 85 mm mortar core gives the wall a permanence that lightweight construction cannot replicate.
- **No visible fasteners, joints, or seams.** Once plastered, the wall is continuous. No one can tell where one panel ends and the next begins. No one can tell there is a steel frame inside. The building looks and feels like traditional masonry construction.

Someone entering a building made with these panels would not know — or care — that the walls were prefabricated in a workshop by a community team. They would see smooth plaster, feel solid walls, and experience a building with the character of handmade construction. That the building also survives earthquakes, lasts 80–100 years, and was built at a fraction of the cost of conventional construction is invisible.

This is the point: affordable and beautiful are not opposites. The same techniques that kept the cost low — lime instead of paint, mortar instead of drywall, bamboo instead of steel studs — are the techniques that produce the most beautiful result.

Build Once — 80–100 Year Lifespan

A house that lasts 15 years is not affordable. It is a debt trap with a roof.

A house that lasts 80–100 years is built once. The family invests once. The community builds once. The materials are consumed once. Everything after that is maintenance — lime wash every 5–10 years, plaster inspection every 15–20 years.

The cost per year of a 100-year building is a fraction of a 15-year building, even if the initial construction costs more.

Community Production

The screw-clamping system was designed specifically for community production:

- **No welding required** during panel assembly (frames are pre-welded in a workshop, but assembly is screws only)
- **No specialized tools** — drill, screwdriver, basic hand tools
- **Self-adjusting** — the clamping strip flexes to accommodate natural thickness variation in guadua strips
- **Teachable in days** — a community team can learn to produce panels with 2–3 days of training

- **Production rate** — a team of 4–6 people with 2 vibration tables produces 3–4 panels per day

Prior Art & Related Systems

This project does not exist in a vacuum. Several systems have explored bamboo-based prefabricated or engineered construction. We build on their work and acknowledge their contributions.

Engineered Bahareque — Sebastian Kaminski (Arup / INBAR)

The most cited academic work on modern bahareque. Kaminski's Design Guide for Engineered Bahareque Housing (published with INBAR and Fundación Redes) provides the structural engineering foundation for cemented bahareque walls — guadua skeleton frames with bamboo mat infill, wire mesh reinforcement, and cement mortar render. His work underpins the Colombian NSR-10 building code for bahareque encementado (Chapter G-1). Over 4,000 homes built across Colombia, Costa Rica, Nepal, Ecuador, Peru, Mexico, El Salvador, and the Philippines.

How this system differs: Kaminski's approach is in-situ construction — walls are built on-site by skilled craftspeople, one layer at a time. The finish depends heavily on workmanship. Our system industrializes this into standardized prefabricated panels: steel T-bar frame (not timber/bamboo), screw-clamped strips (not nailed mats), vibration-table mortar pour (not troweled on-site), and integrated snap-connect electrical. The goal is consistent quality at community-production speed, with a finished wall surface suitable for residential and commercial buildings.

Cement Bamboo Frame Technology (CBFT) — BASE Bahay / Hilti Foundation

The most scaled bamboo construction system in the world: 2,400+ housing units built in the Philippines, Nepal, India, and Sri Lanka. CBFT uses treated bamboo frames with metal connections, flattened bamboo (esterilla) infill, wire mesh, and cement mortar. Panels are prefabricated in workshops. Designed for typhoon winds up to 300 km/h and earthquakes up to magnitude 7–8. Target lifespan: ~60 years. Training provided through the Bamboo Academy.

How this system differs: CBFT uses bamboo/wood as the frame structure. Our system uses galvanized steel T-bar — providing higher axial capacity, longer corrosion-protected lifespan (80–100 years vs. 60), and a consistent clamping surface for the screw system. CBFT does not include integrated electrical. CBFT is proprietary (training-based access); our system is open hardware.

- BASE Bahay — Hilti Foundation
- BASE Builds

Bamboo Composite Shear Walls — Chinese Academic Research

Recent research from Chinese universities has tested sprayed composite mortar walls with original bamboo studs and **diagonal bamboo bracing** — structurally the closest match to our panel's diagonal bracing concept. Quasi-static cyclic loading tests showed good energy dissipation and ductile failure behavior with diagonal strips. The mortar is sprayed rather than vibration-poured, and there is no steel frame.

How this system differs: Our vibration-table pour achieves more complete mortar fill (zero voids) without spray equipment. The steel T-bar frame adds axial load capacity and provides a standardized clamping surface. However, the Chinese research provides the best published validation that diagonal bamboo bracing in mortar composite works structurally — we build on this evidence.

- Diagonal bamboo strips seismic study (ScienceDirect)

Housing NOW / Blue Temple — Myanmar

Modular bamboo housing using interlocking arched bundled bamboo frames. 26 homes built with community labor at ~$1,000 per unit. In March 2025, all 26 survived a magnitude 7.7 earthquake with **zero structural damage** — the most impressive real-world seismic validation of any bamboo housing system to date.

How this system differs: Completely different structural approach (arched bundled bamboo, no mortar, no steel). But their community-production model and proven earthquake survival are powerful validation of bamboo's seismic potential. If bundled bamboo arches survive M7.7, mortar-encased bamboo strips in a steel frame should perform at least as well.

- Housing NOW
- Dezeen — Blue Temple bamboo housing

WikiHouse — UK

Open-source CNC-cut plywood construction system. Creative Commons licensed blueprints on GitHub. Community-assembled. Not bamboo-based, but the closest model for how an open-source construction system can be documented, licensed, and distributed. WikiHouse demonstrates that open-source building is viable.

How this system differs: Different materials (plywood vs. bamboo+mortar+steel), different climate target (temperate vs. tropical), different production method (CNC vs. workshop). But WikiHouse's licensing approach (CC files on GitHub) and community model informed the structure of this repository.

- WikiHouse

Quincha Prefabricada — Peru

Prefabricated quincha panels — sawn timber frames filled with woven round cane in interlocking patterns (no nails needed for infill), plastered with mud-straw mix. Construction manuals publicly available (PREDES). Lima's colonial center was rebuilt in quincha after the 1746 earthquake — these flexible frames survived subsequent earthquakes that destroyed rigid masonry.

How this system differs: Quincha uses timber frames (not steel), mud plaster (not pozzolanic mortar), and woven cane (not screw-clamped strips). No integrated electrical. But quincha prefabricada proves that prefabricated bamboo/cane panels with plaster infill are a viable, earthquake-resistant construction method — a direct ancestral validation of this system's approach.

- Quincha Prefabricada (ResearchGate)
- Manual Quincha Mejorada (PREDES)

Colombian NSR-10 — Bahareque Encementado Standard

Not a system but the regulatory framework: Colombian seismic code NSR-10, Chapter G-1 (formerly E-7), governs cemented bahareque construction. Any building using this panel system in Colombia would reference this standard. The code recognizes guadua frame + bamboo mat + wire mesh + cement mortar as a code-compliant wall system.

- Manual Bahareque Encementado (PDF)

What's New in This System

For clarity, the following features have **no documented equivalent** in any existing system:

Feature	Status
Galvanized steel T-bar frame for bamboo panels	Novel
Screw-clamping system (self-adjusting for thickness variation)	Novel
Vibration table mortar pour on sand bed	Novel
Integrated 12V + 120V snap-connect electrical	Novel
Open hardware license (CERN-OHL-S) for bamboo construction	Novel
80–100 year design lifespan for bamboo-mortar panels	Novel (CBFT targets 60)
)(flex profile from HDPE spacer blocks	Novel

Feature	Status
Diagonal bamboo bracing through corner-notched spacer blocks	Novel combination

This system is an evolution, not an invention from scratch. It combines proven principles from bahareque, quincha, Fachwerk, and Pombaline construction with modern engineering, materials science, and open-source distribution.

Open Source

This system is open source because:

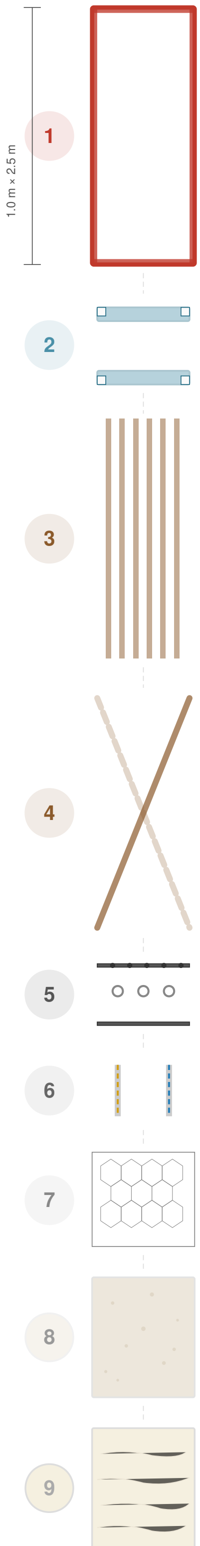
1. Housing is a human right. The knowledge to build safe, dignified shelter should not be a commercial secret.
2. The system gets better with every community that builds. A carpenter in Manizales, a builder in Indonesia, an engineer in Nepal — each can improve the design and share it back.
3. Patents on construction techniques create artificial scarcity of knowledge. Publishing the design as open hardware creates permanent prior art that no one can close.
4. The best construction systems in history spread through open sharing — vernacular architecture evolved over centuries through exactly this process. We're just doing it with version control.

This project is licensed under CERN-OHL-S v2 (hardware) and CC BY-SA 4.0 (documentation). This means all derivatives must remain open. The knowledge stays free, forever.

Panel Anatomy

Panel Exploded View — 1.0 m × 2.5 m

Components separated vertically · Numbers correspond to assembly order



T-bar frame
30×30×3 mm galvanized steel
Welded corners, drilled clamping holes

HDPE spacer blocks
30×30 mm full width, corner notches
Space strips 30 mm from web at top/bottom

Bamboo vertical strips
~20 mm wide, gaps for mortar penetration
~27 per side, screw-clamped to web
Borate-treated for insect/fungal protection

Diagonal bracing
60×20 mm, 1 per side from opposing corners
Forms X pattern · 3–5× seismic improvement
Solid = front · Dashed = rear side

Clamping strips + wire ties
2 mm steel strips, screws every ~70 mm
Wire ties at every diagonal-vertical crossing

PVC conduit + cables
12V lighting (gold) — 6× E10 sockets
120V mains (blue) — snap connectors

Chicken wire mesh
Galvanized hex 25 mm, both faces
+ optional aluminum insect mesh beneath

Cement mortar
1:4 cement:sand + pozzolanic admixture
Vibration table pour · ~85 mm total thickness
Fills center completely through strip gaps

Lime plaster + lime wash
3–5 mm hand-troweled, both faces (on-site)
Breathable, antifungal, self-healing
Lime wash: soft matte finish, each stroke unique

Color Key

Steel frame · HDPE · Bamboo · Clamps/wire · Conduit · Mesh · Mortar · Plaster

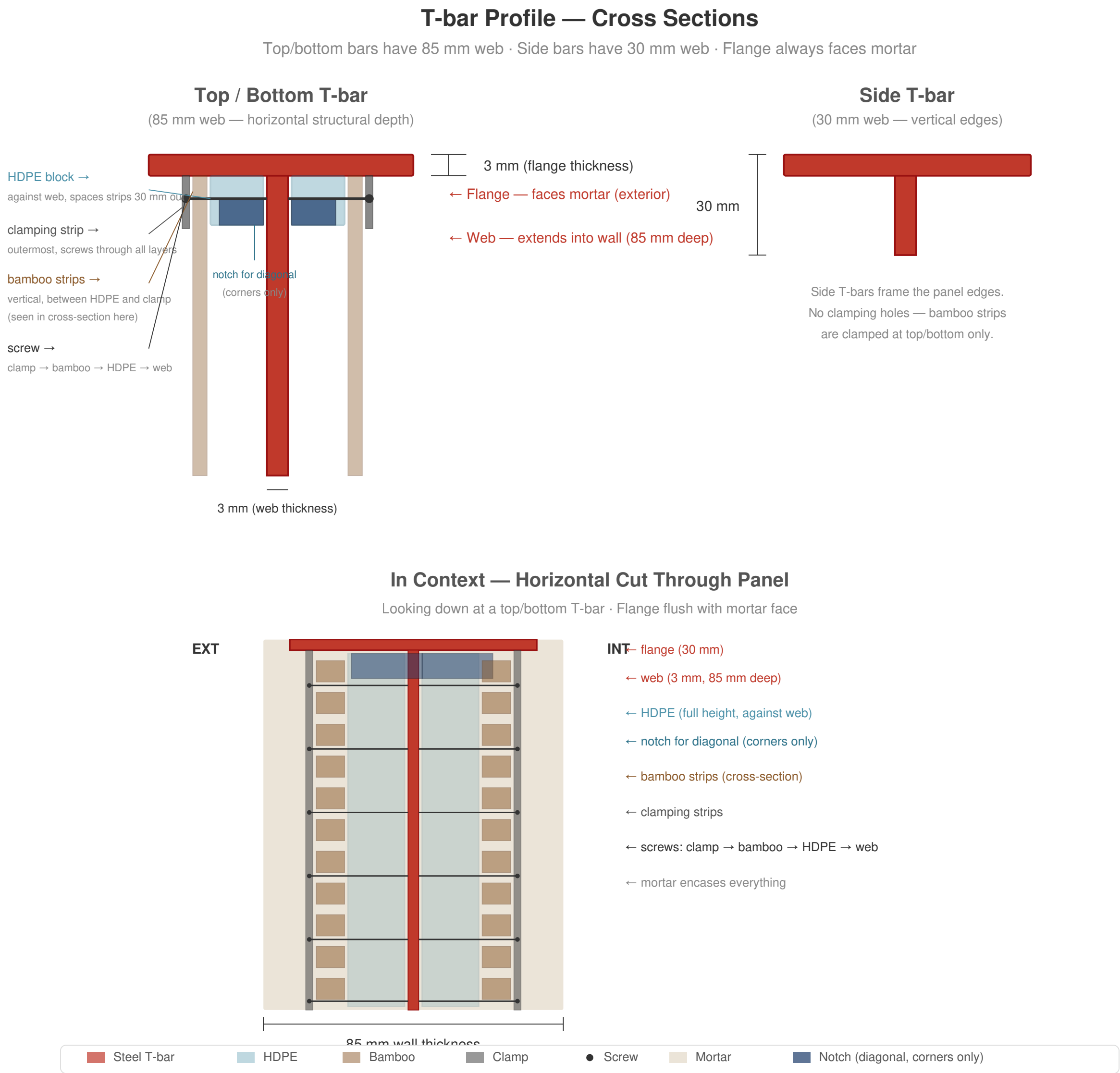
Panel weight: ~155 kg (2.5 m) / ~183 kg (3.0 m) · Carried by 3–5 people

Overview

Every wall panel is **1.0 m wide × 2.5 m tall** (vertical orientation), weighs approximately **155 kg**, and contains integrated structure, insulation, and electrical systems. One size. Four variants. Fully prefabricated in a workshop, transported to site, bolted into a steel building frame.

> ***Scalability:*** *The 2.5 m height suits standard residential ceiling heights worldwide. The system scales to any height — 3.0 m for high ceilings, 2.7 m for commercial spaces, 2.0 m for partitions. Only the vertical T-bars and bamboo strips change length. The frame jig, clamping system, mortar process, and electrical layout remain identical.*

Frame: T-bar 30×30×3 mm



The panel frame is hot-dip galvanized steel T-bar profile:

- **Flange:** 30 mm wide × 3 mm thick — faces outward, flush with the mortar surface
- **Web:** 3 mm thick — extends inward, provides structural depth and clamping surface
- **Top/bottom T-bars:** 85 mm web height (provides horizontal structural depth)

- **Side T-bars:** 30 mm web height
- **Corners:** Welded in a jig — all frames identical
- **Clamping holes:** Drilled every ~70 mm along the top and bottom webs (one hole per two bamboo strips)

The frame is the structural backbone. Everything else attaches to it.

HDPE Spacer Blocks

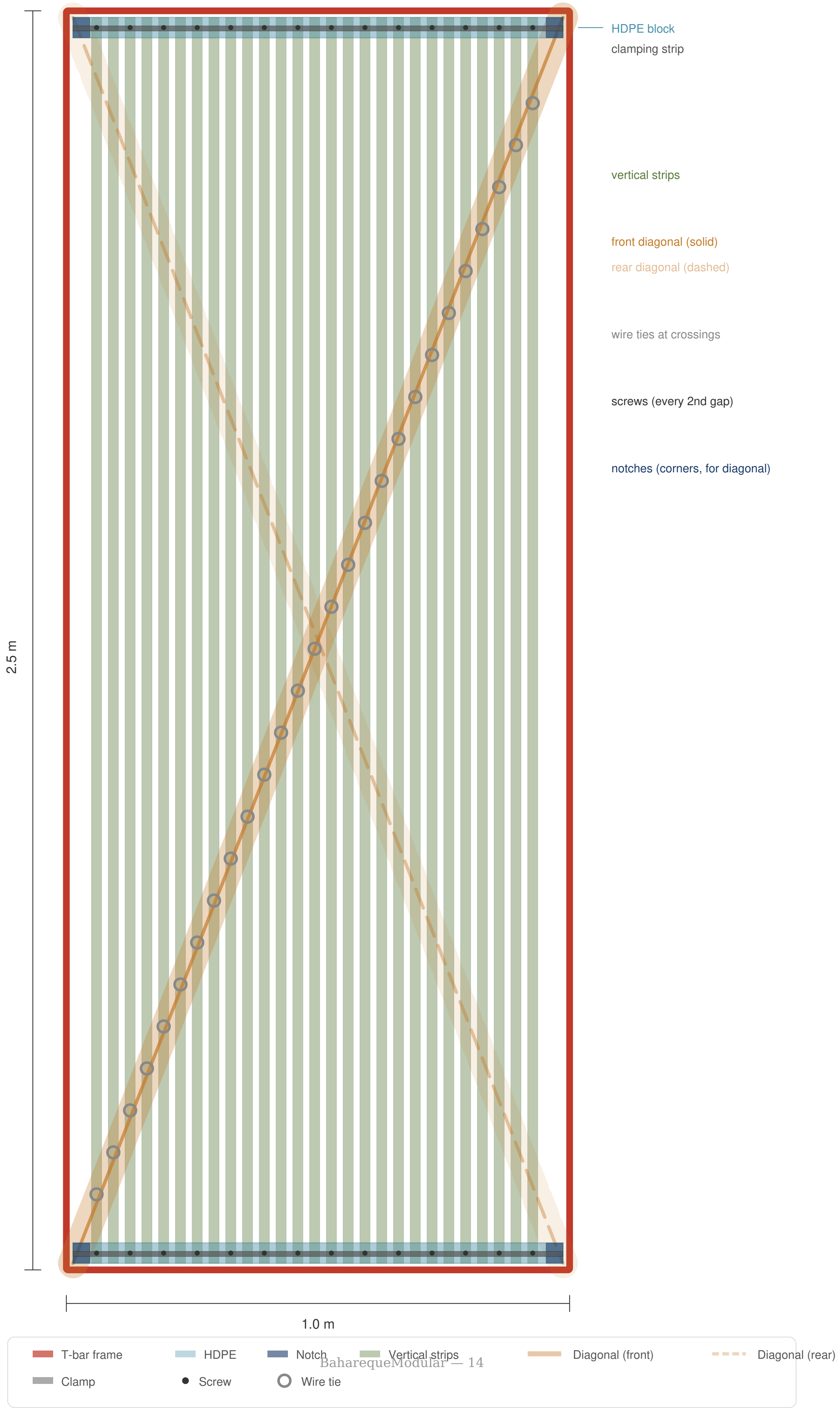
- **Size:** 30 × 30 mm cross-section, full 1 m width
- **Position:** Mounted on top and bottom T-bar webs (2 per panel)
- **Function:** Space bamboo strips 30 mm away from the web at top and bottom edges
- **Corner notches:** 10 × 10 mm cutouts at each corner anchor the diagonal strips at web level
- **Material:** Recycled HDPE (from sheet or pipe). Zero rot, zero corrosion, dimensionally stable

Guadua Vertical Strips

- **Material:** Borate-treated Guadua angustifolia, split into strips
- **Dimensions:** ~20 mm wide × 2,500 mm long
- **Spacing:** ~20 mm gaps between strips (mortar penetration)
- **Quantity:** ~27 strips per side, ~54 total per panel
- **Attachment:** Screw-clamped to the T-bar web at top and bottom via clamping strips

Face View

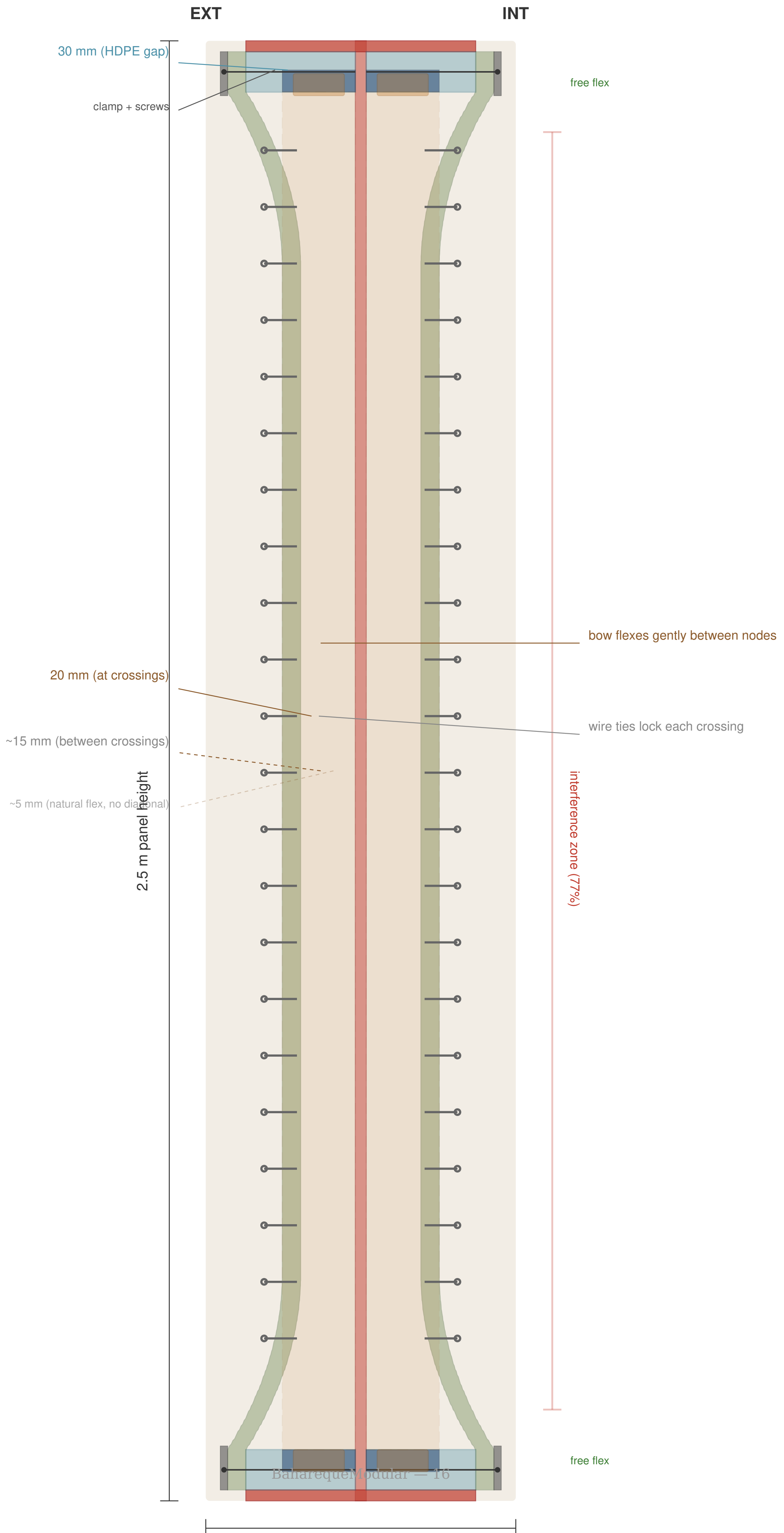
Front side, 1.0 m × 2.5 m



The)(Profile

Side Profile

Cross-section through wall depth (not to scale)



At the top and bottom, the HDPE blocks hold the strips 30 mm from the web. At mid-height, the strips flex naturally inward toward the web — creating a **)(** cross-section profile. This is not a defect; it is the design:

- The mortar fills the varying gap, creating a natural arch shape
- The arch resists out-of-plane forces (wind, impact)
- The variable thickness mortar locks the strips mechanically

Bamboo Diagonal Strips

- **Dimensions:** 60 × 20 mm, ~2,690 mm long (corner to corner diagonal)
- **Quantity:** 1 per side, from opposing corners (forms an X when viewed from front)
- **Position:** Runs at web level, threaded through the HDPE block corner notches
- **Pre-tensioned:** Pulled tight before securing
- **Function:** Converts seismic shear forces to tension in the diagonal. Provides **3–5× improvement in racking resistance** over panels without diagonals.

Wire Ties

Galvanized wire ties at every diagonal-vertical crossing (~8–10 per side). These lock the vertical and diagonal strips into a rigid grid, distributing point loads across the entire panel face and creating a ductile failure mode.

PVC Conduit

- **Size:** 16 mm standard electrical conduit
- **Position:** Between bamboo strips, against the web
- **Function:** Protects 12V and 120V cables from mortar and screw pressure. Allows cable replacement by pulling new wire through without opening the panel.

Electrical Systems

Each panel contains two independent circuits:

12V Lighting

- 2-core cable in PVC conduit
- 6× E10 screw sockets (3 per side) near the top of the panel
- Warm incandescent or LED bulbs (0.5–1W each) — wall wash lighting
- 2-pin snap connectors at both vertical edges

120V Mains (per variant)

- 3-core cable (L + N + G) in PVC conduit
- 3-pin snap connectors at both vertical edges
- When panels are bolted adjacent, connectors click together = continuous circuit

Panel Variants

Type	Share	Contents
P — Pass-through	~60%	12V lighting + 120V pass-through cable. No outlets.
O — Outlet	~18%	12V lighting + duplex outlet at ~40 cm height
S — Switch + Outlet	~9%	12V lighting + switch at ~120 cm + outlet at ~40 cm
W — Water + Outlet	~13%	12V lighting + outlet + cold/hot water risers + grey water drain

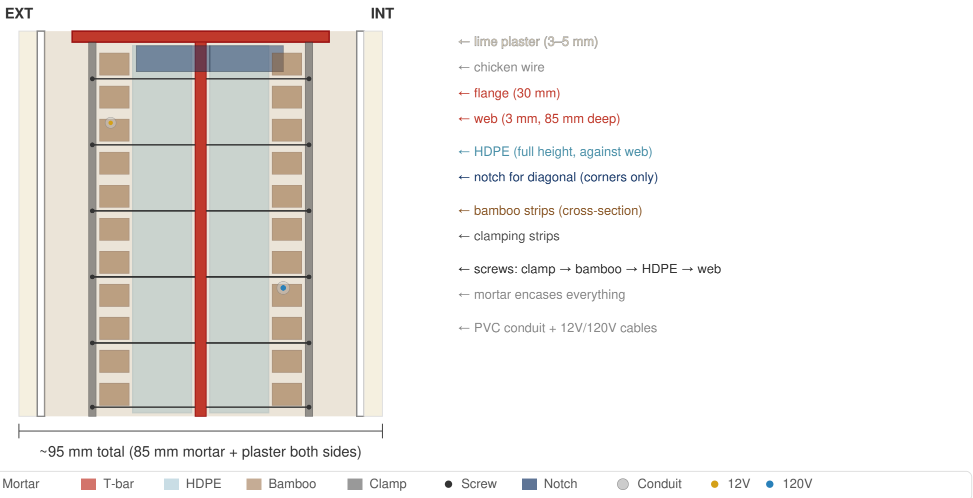
All variants share the same frame, same bamboo infill, same mortar. Only the embedded services differ.

Mortar

- **Mix:** 1:4 Portland cement : clean river sand
- **Additives:** Polypropylene fiber (6–12 mm) for curing-phase crack prevention + pozzolanic admixture (volcanic ash, rice husk ash, or metakaolin)
- **Application:** Poured on vibration table (see Construction Process)
- **Total wall thickness:** ~85 mm (mortar + guadua + mortar)
- **The pozzolan** reduces mortar pH over time, slowing degradation of embedded bamboo

Panel Cross Section — Horizontal Cut at Top/Bottom T-bar

Looking down · Shows full wall assembly · Exterior left, interior right



Finish Layers (applied on-site after installation)

1. **Chicken wire mesh** — galvanized hexagonal mesh (25 mm opening), stapled to both faces. Provides mechanical key for mortar/plaster adhesion.
2. **Fine aluminum mesh** (optional) — standard insect/dust screen, 1–1.5 mm opening. Blocks insects, fine dust, and pollen from entering through the mortar matrix. As a secondary property, the aluminum layer also provides measurable radio-frequency attenuation.
3. **Lime plaster** — 3–5 mm, hand-troweled. Breathable, antifungal, self-healing. Optional: chopped dried grass fiber for crack resistance.
4. **Lime wash** — cal + water, brush-applied. Soft matte finish. Each brush stroke unique.

Weight Breakdown (approximate)

Component	Weight
Steel frame	~18 kg
HDPE blocks	~1 kg
Bamboo strips + diagonals	~12 kg
Mortar (85 mm × 1 m × 2.5 m)	~120 kg
Wire, mesh, conduit, cables	~6 kg
Total	**~155 kg**

Carried by 3–4 people using a simple frame jig.

Materials

Bill of Materials (per panel)

Material	Specification	Quantity	Notes
T-bar steel	30×30×3 mm, hot-dip galvanized	2× 1.0 m + 2× 2.5 m = 7.0 m	One welding jig for all panels
Clamping strips	2 mm flat steel, galvanized, matching holes	4× 1.0 m = 4.0 m	Top + bottom, both sides
Screws	Stainless steel, self-tapping	~28 per panel	Through clamping strip → bamboo → HDPE → web (every ~70 mm, one per two strips)
HDPE blocks	30×30 mm cross-section, recycled	2× 1.0 m = 2.0 m	Top + bottom T-bars
Bamboo strips (vertical)	Borate-treated, ~20 mm wide	~54 strips × 2.5 m	~27 per side
Bamboo strip (diagonal)	Borate-treated, 60×20 mm	2× ~2.69 m	1 per side, opposing corners
Galvanized wire	Tie wire	~5 m	~16–20 ties at crossings
PVC conduit	16 mm electrical	~5 m	Cable protection
12V cable	2-core	~6 m	Lighting circuit
120V cable	3-core (L+N+G)	~6 m	Mains circuit
E10 sockets	Screw-base lamp holders	6	3 per side
Snap connectors	2-pin (12V) + 3-pin (120V)	2 + 2	Both vertical edges
Chicken wire	Galvanized hex mesh, 25 mm	~5.5 m^2	Both faces + overlap
Cement mortar	1:4 cement:sand + additives	~0.21 m^3	~120 kg wet weight
PP fiber	6–12 mm chopped	~200 g	Curing crack prevention
Pozzolanic admixture	Volcanic ash / rice husk ash / metakaolin	~2 kg	pH reduction, durability

Lime plaster and lime wash are applied on-site after panel installation — not part of the panel itself.

Material Details

Structural Bamboo

The system uses split structural bamboo strips as the primary infill material. The reference species is **Guadua angustifolia** (known as *guadua* across Latin America, *caña guadua* in Ecuador, *tacuara* in some regions) — the strongest structural bamboo in the Americas. However, any large-diameter structural bamboo species can be adapted.

Key properties of Guadua angustifolia:

- **Tensile strength:** 150–200 MPa (comparable to mild steel)
- **Growth:** Reaches harvest diameter (10–15 cm) in 4–5 years
- **Range:** 0–2,000 m altitude across tropical Americas (Colombia, Ecuador, Central America)

- **Harvest:** Cut at 4–6 years (maximum lignification). Younger culms are too soft; older ones become brittle.
- **Treatment:** Borate solution (borax + boric acid) by modified Boucherie method or soaking. Prevents insect attack and fungal decay. Non-toxic, effective for 20+ years when encased in mortar.

Alternatives for Other Regions

Species	Region	Notes
Dendrocalamus asper	Southeast Asia	Large diameter, strong. Direct substitute.
Bambusa bambos	South/Southeast Asia	Thorny, strong. Requires wider strips.
Phyllostachys edulis (Moso)	China, temperate zones	Smaller diameter. Use wider, thinner strips.
Phyllostachys bambusoides	Japan, temperate zones	Good strength. Smaller culms.
Guadua chacoensis	Argentina, Paraguay	Close relative of G. angustifolia.

The system works with any structural bamboo that can be split into strips. Adjust strip width and spacing for the available species.

Steel T-bar

- **Profile:** 30×30×3 mm (30 mm flange, 30 mm or 85 mm web, 3 mm thickness)
- **Galvanizing:** Hot-dip per ISO 1461, minimum 85 μm zinc coating
- **Lifespan:** 80–100 years in non-coastal tropical environments at the specified zinc thickness
- **Sourcing:** Any structural steel supplier. T-bar is a standard profile worldwide.
- **Alternative:** If T-bar is unavailable, weld an angle (30×30×3) to flat bar (30×3) to create the profile. Same function, slightly more welding.

HDPE Spacer Blocks

- **Source:** Recycled HDPE cutting boards, pipe sections, or sheet stock
- **Properties:** Zero water absorption, rot-proof, dimensionally stable, easy to machine
- **Machining:** Table saw or band saw to cut to size. Router or chisel for corner notches.

Alternatives to HDPE

Material	Pros	Cons
Recycled HDPE (preferred)	Rot-proof, zero water absorption, widely available from waste stream	Requires cutting tools
Marine-grade hardwood (teak, ipe, greenheart)	Strong, naturally rot-resistant, easy to machine	Expensive, still degrades over decades in moist environments
Treated softwood (CCA or borate-treated pine)	Cheap, widely available	Shorter lifespan (20–40 years), may need replacement
Dense rubber blocks (recycled tire rubber)	Rot-proof, shock-absorbing, free from waste tires	Harder to machine precisely, may compress over time
Concrete/mortar blocks	Zero cost, cast in molds, permanent	Brittle, harder to notch for diagonals, heavy
3D-printed blocks (PETG or ASA)	Precise geometry including notches, reproducible	Requires 3D printer, UV degradation if exposed

The critical requirement is: the block must maintain its 30 mm dimension over the building's lifetime to preserve the)(strip profile. HDPE meets this best, but any material that resists moisture and compression will work.

Mortar

- **Base mix:** 1 part Portland cement Type I : 4 parts clean river sand (well-graded, 0–4 mm)
- **PP fiber:** 200 g per panel (~1 kg per m^3). Prevents plastic shrinkage cracks during curing.
- **Pozzolanic admixture:** 5–10% cement replacement.
- **Volcanic ash** — available in Andean regions, naturally pozzolanic
- **Rice husk ash (RHA)** — available in rice-growing regions (Asia, Latin America)
- **Metakaolin** — processed kaolin clay, available globally but more expensive
- **Function:** Reacts with calcium hydroxide in cement, reducing pH from ~12.5 to ~10–11 over time. This slows degradation of embedded guadua.
- **Water:** Clean, potable. W/C ratio ~0.45–0.50 for pourable consistency on vibration table.

Lime Plaster (on-site)

- **Type:** Hydraulic lime (NHL 3.5 or NHL 5) or slaked lime (putty lime) + sand
- **Mix:** 1:2.5 to 1:3 lime:sand
- **Optional fiber:** Chopped dried grass (star grass, sisal, hemp) at ~1 cm length. Adds tensile strength and crack resistance. Natural cellulose lasts 30+ years in lime (pH ~10, much less aggressive than cement).
- **Application:** Hand-troweled, 3–5 mm. The texture is the beauty — machine-perfect is not the goal.

Lime Wash (on-site)

- Slaked lime + water, brush-applied in thin coats
- Penetrates and bonds chemically with lime plaster (carbonation)
- Reapply every 5–10 years for maintenance
- Can be tinted with natural pigments (earth oxides) for color

Chicken Wire

- Galvanized hexagonal mesh, 25 mm opening, 0.7–0.9 mm wire
- Stapled or tied to panel face before mortar application
- Provides mechanical key for mortar adhesion
- Both faces

Fine Aluminum Mesh (optional)

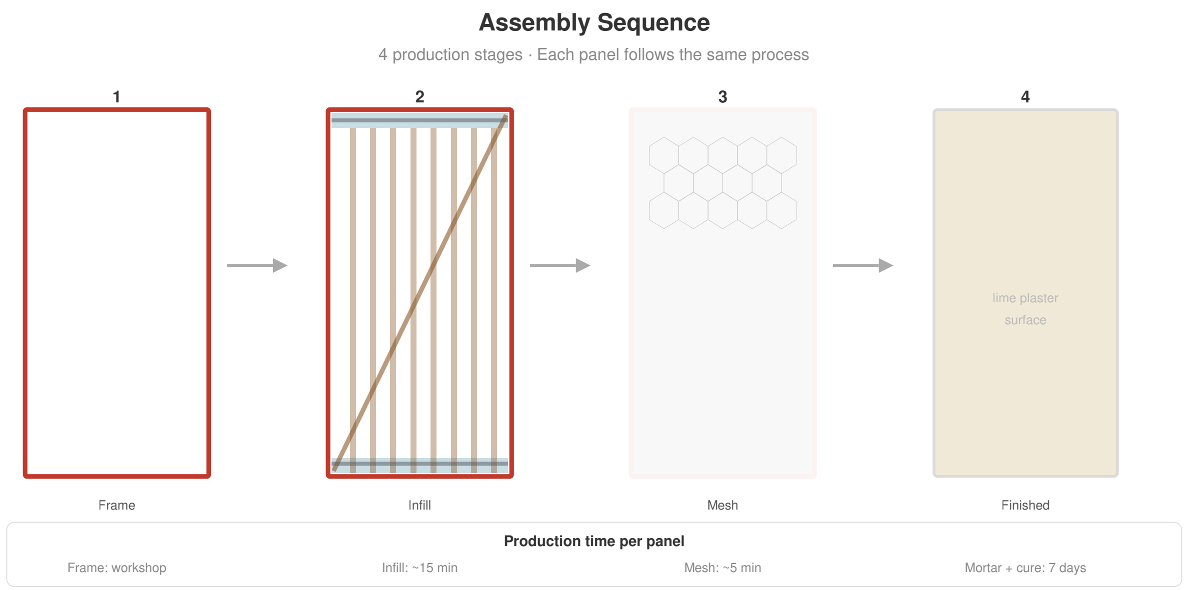
- Standard insect/dust screen, 1–1.5 mm opening, aluminum
- Placed under chicken wire on both faces
- Blocks insects, fine dust, and pollen from entering through the mortar matrix
- As a secondary property, also provides measurable radio-frequency attenuation
- Not required for structural function — add based on local insect/dust conditions

Construction Process

Workshop Setup

All panels are built in a covered workshop — weather-independent, quality-controlled, assembly-line efficient. Minimum workshop requirements:

- Covered area: ~6 × 4 m (roof, open sides acceptable)
- Welding station (frame fabrication only)
- Drill press or hand drill
- 2× vibration tables (~1.2 × 2.7 m each)
- Sand supply for bed
- Mortar mixer (electric or hand)
- Basic hand tools: screwdrivers, wire cutters, pliers, staple gun



Production Steps

Step 1: Frame Fabrication

1. Cut T-bar to length: 2× 1.0 m (top/bottom, 85 mm web) + 2× 2.5 m (sides, 30 mm web)
2. Weld corners in a jig — all frames identical
3. Drill clamping holes every ~70 mm along the top and bottom T-bar webs (one per two bamboo strips, ~14 holes per bar, ~28 total)
4. Hot-dip galvanize the completed frame (batch process — send 20–50 frames at once)

All frames are identical regardless of panel variant. The jig ensures consistent dimensions. One jig, one frame, forever.

Step 2: HDPE Blocks

1. Cut HDPE stock to 30 × 30 mm × 1,000 mm (2 per panel)

2. Cut corner notches: 10 × 10 mm at each end (4 notches per block, 8 per panel)
3. Mount blocks on top and bottom T-bar webs using screws or adhesive

Step 3: Electrical

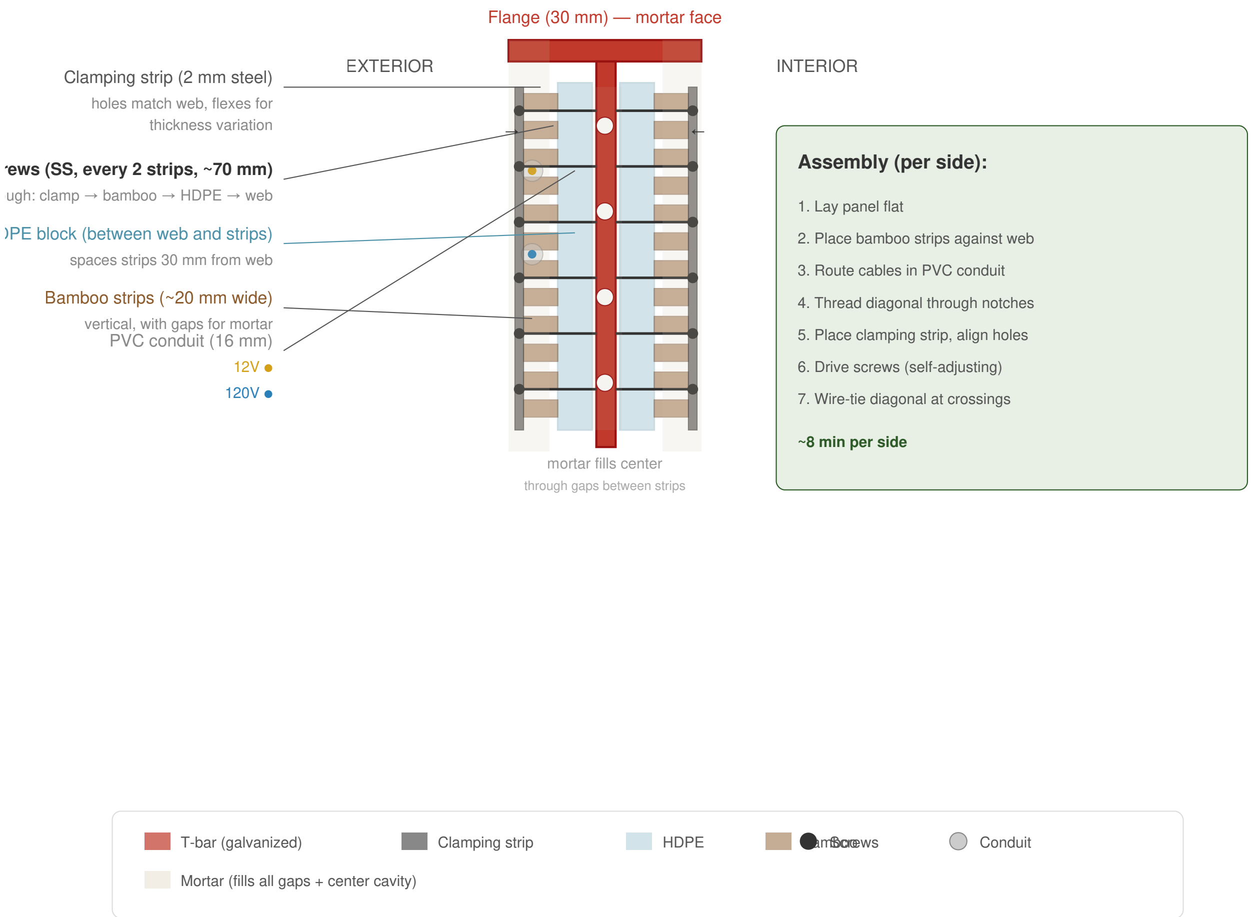
1. Route 12V cable through PVC conduit
2. Route 120V cable through separate PVC conduit
3. Mount E10 sockets (3 near top, each side)
4. For Type O/S: mount outlet box, wire pigtail to main cable
5. For Type S: mount switch box at 120 cm height
6. For Type W: mount water risers (CPVC/PEX supply, PVC drain) with capped stub-outs
7. Attach snap connectors at both vertical edges (2-pin for 12V, 3-pin for 120V)

Step 4: Bamboo Strips (Side 1)

1. Place panel frame flat on work table, side 1 up
2. Lay vertical bamboo strips against the web, between the HDPE blocks
3. Leave ~20 mm gaps between strips for mortar penetration
4. Place PVC conduit with cables between strips, against web
5. Thread diagonal strip through bottom-left HDPE notch, across to top-right notch (at web level)
6. Pre-tension diagonal and secure at both ends

Screw-Clamping System

Cross-section looking down at 1 m width · bamboo strips clamped between web and steel strip



Step 5: Screw-Clamping (Side 1)

1. Place steel clamping strip (2 mm × 30 mm × 1,000 mm) over the bamboo strips at the top T-bar
2. Align holes in clamping strip with holes in T-bar web
3. Drive stainless steel screws through: clamping strip → bamboo strips → web holes
4. The clamping strip flexes to accommodate natural thickness variation — self-adjusting
5. Repeat at bottom T-bar
6. Wire-tie diagonal to each vertical strip at crossings (~8–10 ties)

Time: ~5 minutes per side for clamping, ~3 minutes for wire ties

Step 6: Flip and Repeat (Side 2)

1. Flip panel
2. Repeat steps 4–5 on the other side
3. Diagonal on side 2 runs from top-left to bottom-right (opposing the side 1 diagonal — together they form an X)

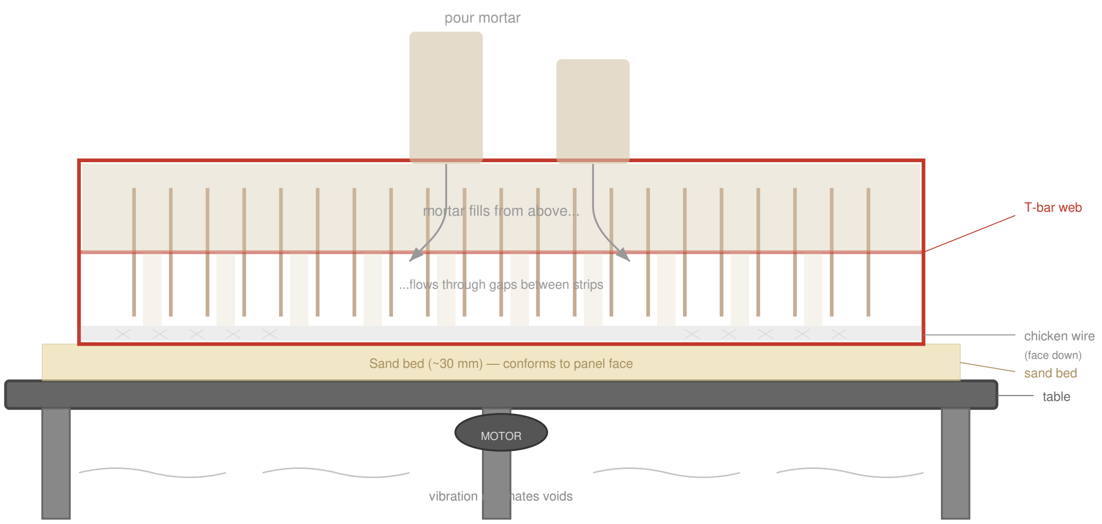
Step 7: Mesh

1. Lay chicken wire over side 1, extending 20 mm past frame edges (for overlap with adjacent panels)
2. Staple or wire-tie to frame and bamboo strips
3. If using aluminum insect mesh: place under chicken wire
4. Flip and repeat on side 2

Step 8: Mortar Pour

Mortar Pour — Vibration Table Method

Side view · Panel face-down on sand bed · Mortar poured from above



Why vibration table on sand bed:

- Gravity + vibration = complete fill, zero voids
- Sand conforms to panel — no mold needed
- No spray equipment (cheaper)
- One-sided pour fills both sides

This is the key innovation in application method:

1. **Prepare vibration table:** Level surface, motor mounted underneath

2. **Spread sand bed:** ~30 mm of clean dry sand on the table surface. The sand conforms to any panel face irregularities and prevents mortar adhesion to the table.
3. **Place panel face-down** on the sand bed (chicken wire side touching sand)
4. **Mix mortar:** 1:4 cement:sand + PP fiber + pozzolanic admixture, W/C ~0.45–0.50 (pourable but not liquid)
5. **Pour mortar** over the exposed back of the panel, filling between strips and covering the mesh
6. **Start vibration:** Run table for 30–60 seconds. The vibration:
7. Drives mortar through the gaps between strips
8. Fills the center cavity completely
9. Eliminates voids and air pockets
10. Mortar penetrates to the front face mesh (now resting on sand)
11. **Screed** excess mortar level with the frame edges
12. **Trowel** smooth (this becomes the interior face)

> ***Vibration table build guide:*** *A DIY vibration table design (motor, platform, springs) will be published in this repository once the design has been tested and proven safe in practice. Until then, any flat platform with a concrete vibrator clamped to the edge will work for initial experiments.*

Why vibration table on sand bed? - Gravity + vibration = complete fill, no voids - Sand bed = no mold needed, sand conforms to panel shape - No spraying equipment needed (cheaper than spray mortar) - One-sided pour fills both sides — mortar flows through strip gaps - Consistent quality — vibration eliminates human error

Step 9: Curing

1. Cover panel with plastic sheet to retain moisture
2. Keep moist for minimum 7 days (mist spray or damp burlap)
3. Full strength at 28 days
4. Panels can be stacked vertically (edge-on) after 7 days to save space
5. Mark each panel with variant type and production date

Production rate: 3–4 panels per day with 2 vibration tables running in parallel, team of 4–6 people. Active work per panel: ~15–20 minutes. Passive curing: 7–28 days (panels cure while new ones are produced).

Step 10: Transport and Installation

1. Transport panels to building site using simple A-frame cart or truck
2. Bolt panels into steel building frame at column positions (pre-welded bolt tabs on columns)
3. Adjacent panels touch mortar-to-mortar
4. Snap-connect electrical between adjacent panels (12V + 120V)
5. Test circuits before finishing

Step 11: On-Site Finishing

1. Fill any joints between panels with mortar
2. Apply chicken wire mesh over joints (if not already overlapping from panel edges)
3. Apply lime plaster: 3–5 mm, hand-troweled, both faces
4. Allow to cure (keep moist for 3–5 days)
5. Apply lime wash: 2–3 thin coats with brush

The finished wall is seamless — no visible joints, no visible steel, no visible screws. Just smooth lime plaster with the texture of a hand-made wall.

Quality Checklist

- [ ] Frame dimensions within ±2 mm
- [ ] All clamping holes drilled and aligned
- [ ] Bamboo strips borate-treated (green/blue tint visible)
- [ ] Diagonal pre-tensioned (should ring when tapped)
- [ ] All wire ties tight
- [ ] Electrical circuits tested before mortar (12V and 120V)
- [ ] Mortar consistency correct (slump test)
- [ ] Vibration run for full 30–60 seconds
- [ ] No visible voids on pour face after vibration
- [ ] Panel marked with type and date
- [ ] Cured minimum 7 days before handling

Structural Performance

System Architecture

This panel system was designed to work **with a steel column-and-beam frame** — the configuration we recommend for multi-story buildings in seismic zones. In that configuration, the steel frame carries gravity and seismic loads while the panels provide lateral bracing, weather enclosure, thermal mass, and integrated services.

However, the panels are **structurally capable of working without a steel frame**. With a conservative axial capacity of ~4,000 kg per panel (two side T-bars), a row of panels can comfortably carry a second floor and roof without an independent steel structure. A simple timber or bamboo ring beam at the top connects the panels and distributes the roof load.

Two configurations:

Configuration	Use case	Panel role
With steel frame (recommended)	Multi-story, high seismic zones, commercial/ institutional	Infill — panels bolt into frame, steel carries primary loads
Without steel frame (simpler, lower cost)	Single story + loft/mezzanine, moderate seismic zones, residential	Load-bearing — panels carry roof and upper floor directly

The load capacity estimates below apply to both configurations. The steel frame adds redundancy and moment resistance, but is not required for the panels to perform structurally.

Seismic Performance

Diagonal Bracing

Each panel contains two diagonal bamboo strips (one per side, opposing corners), forming an X when viewed from the front. Under seismic loading:

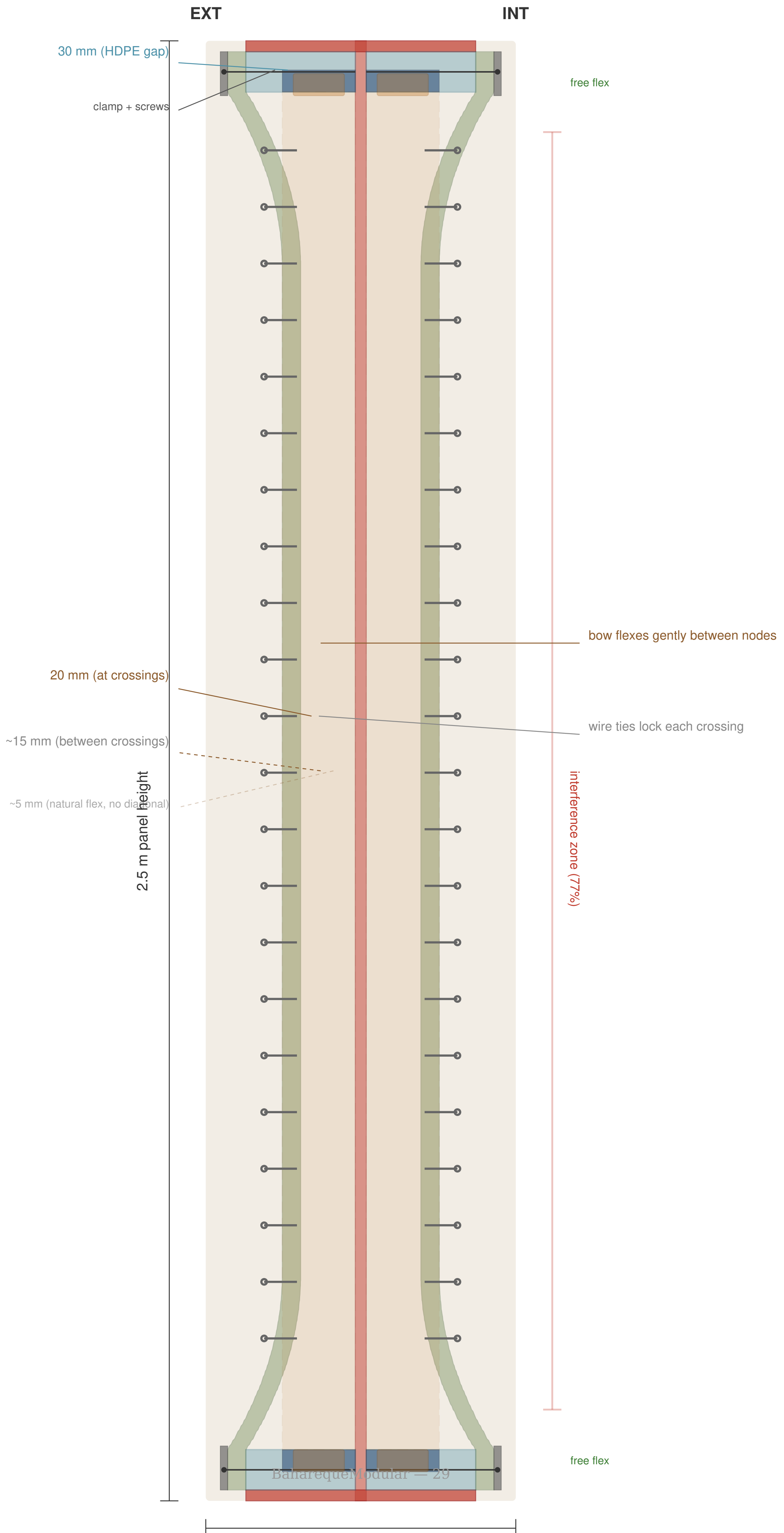
1. **Shear → tension conversion:** When the building frame racks sideways, one diagonal goes into tension while the opposite goes into compression. The bamboo diagonal in tension resists the racking force efficiently — bamboo has excellent tensile strength (150–200 MPa for Guadua angustifolia).
2. **Pre-tension:** Diagonals are installed with slight pre-tension, eliminating slack and ensuring immediate engagement under load.
3. **Wire tie grid:** Galvanized wire ties at every diagonal-vertical crossing (~16–20 ties total) distribute the diagonal force across the full panel face, preventing localized failure.

Estimated improvement: 3–5× racking resistance compared to panels without diagonal bracing.

The)(Profile and Diagonal Interference

Side Profile

Cross-section through wall depth (not to scale)



The HDPE blocks hold bamboo strips 30 mm from the web at top and bottom. Without the diagonal, strips would flex inward to ~5 mm from the web at mid-height, creating a smooth)(curve.

In practice, the diagonal modifies this profile. The diagonal strip (20 mm thick) runs at web level behind the vertical strips. At each of the ~22 crossing points in the middle 77% of the panel, the diagonal holds the vertical strip at 20 mm from the web — preventing full flex. Between crossings, strips flex slightly more. The result is a **ribbed)(profile**: a series of rigid nodes (at wire-tied crossings) with slight flex between them.

This interference is **by design and beneficial:**

- **Out-of-plane resistance:** The mortar between the strip layers still forms an arch, but with more pronounced ribs at each crossing — creating more mechanical interlock points than a smooth curve.
- **Mechanical interlock:** The varying mortar thickness at and between crossings locks the bamboo strips more securely than a uniform gap would.
- **Buckling resistance:** Strips held at 20 mm from the web (vs 5 mm with natural flex) contribute **7× more** to the wall's moment of inertia. This increases the critical buckling load by ~10% — meaning the diagonal improves not just racking resistance but also vertical load capacity.

Zone	Panel height	Gap to web	Behavior
HDPE block zone (top/bottom)	0–12% and 88–100%	30 mm	Free flex, no interference
Interference zone (middle)	12–88%	20 mm (held by diagonal)	Diagonal acts as spacer, wire ties lock crossings

For builders: The diagonal strip is installed first (threaded through HDPE notches at web level), then the vertical strips are placed against it. At crossing points, the vertical strip rests on the diagonal — they touch, and the wire tie locks them together. No notching, cutting, or special treatment is needed. The mortar fills all gaps during the vibration table pour.

Ductile Failure Mode

The combination of bamboo strips, wire ties, and mortar creates a composite that fails gradually:

1. First: mortar cracks (visible warning)
2. Then: wire ties stretch (energy absorption)
3. Then: bamboo strips deform (high ductility)
4. The steel frame remains intact throughout — the building does not collapse

This is fundamentally different from unreinforced masonry, which fails brittlely (sudden collapse with no warning).

Fire Resistance

- **Mortar encasement:** The bamboo strips are fully encased in mortar (~85 mm wall thickness). Mortar is non-combustible and acts as thermal insulation for the bamboo.
- **Bamboo charring:** Even if fire penetrates the mortar, bamboo chars slowly and predictably (similar to timber construction). The steel frame behind maintains structural integrity.
- **Lime plaster finish:** Non-combustible. Provides additional fire protection.
- **No exposed bamboo:** Once installed and finished, no bamboo is visible or accessible to flame.

Thermal Performance

- **Thermal mass:** ~85 mm of mortar provides significant thermal mass — absorbs heat during the day, releases at night. Reduces temperature swings in tropical climates.
- **Breathability:** Lime plaster is vapor-permeable. Moisture passes through the wall rather than getting trapped, preventing condensation and mold.

- **No insulation gap:** The system does not include a thermal insulation layer. In tropical climates (the primary target), thermal mass is more valuable than insulation. For temperate climates, an external insulation layer can be added outside the lime plaster.

Acoustic Performance

The dense mortar core provides good sound attenuation:

- **Estimated STC (Sound Transmission Class):** 40–45 for a single panel wall
- **With lime plaster on both sides:** STC 42–48
- **Comparison:** Standard concrete block wall: STC 40–45. The panel system is comparable.

Load Capacity

> ***Disclaimer:*** *The values below are conservative analytical estimates based on material properties and geometry. They are NOT based on laboratory testing. Do not use these values for structural engineering without independent verification. Actual performance depends on material quality, workmanship, and connection details. See* *Testing Needed* *below.*

Panel Self-Weight

Component	Weight
Steel frame (T-bar 30×30×3, 7 m total)	~18 kg
Mortar (85 mm × 1.0 m × 2.5 m, density ~2,000 kg/m³)	~120 kg
Bamboo strips + diagonals	~10 kg
HDPE blocks, wire, mesh, conduit, cables	~7 kg
Total panel weight	**~155 kg**
Weight per m² of wall	**~62 kg/m²**

For comparison: 150 mm concrete block wall ≈ 180 kg/m². The panel system is roughly one-third the weight of conventional masonry per unit area.

Axial (Vertical) Load

Each panel has two vertical side T-bars that can carry significant compressive loads:

Element	Cross-section	Conservative axial capacity
Side T-bar (30×30×3 mm)	A ≈ 171 mm²	~20 kN per T-bar (at σ = 120 MPa, ~50% of yield)
Two side T-bars per panel	2 × 171 mm²	~40 kN per panel (~4,000 kg)

What 4,000 kg per panel means in practice:

A typical single-story house with a loft/mezzanine and terracotta tile roof imposes roughly 200–400 kg of load per linear meter of wall (roof + ceiling + mezzanine floor + live load). A single 1 m panel can carry **10–20× this requirement**.

Even a conservative 10-panel wall (10 m) carries a combined 40,000 kg — far more than any realistic residential roof and upper floor. **The panels do not need a steel frame to carry a second floor and roof.**

When used **with a steel frame**, this axial capacity is pure reserve margin — the steel carries the gravity loads. When used **without a steel frame**, the panels are the load-bearing structure. In that case, a timber or bamboo ring beam at the top of the wall distributes the roof load evenly across all panels and ties them together.

Buckling is not a concern at these load levels: the 85 mm mortar wall thickness provides substantial lateral stiffness to the T-bar, and the panel height-to-thickness ratio (~2,500/85 ≈ 29) is well within accepted slenderness limits for reinforced walls. Additionally, the diagonal bracing holds bamboo strips at 20 mm from the web (instead of ~5 mm with natural flex), which increases the bamboo's contribution to the moment of inertia by ~7× — improving the critical buckling load by approximately 10%. The diagonal therefore improves both racking resistance and buckling resistance simultaneously.

Lateral (In-Plane / Racking) Load

This is the panel's primary structural contribution — resisting horizontal forces from wind and earthquakes.

Without diagonal bracing (mortar + vertical strips only):

Conservative estimate based on mortar shear strength ($\tau \approx 0.3$ MPa for 1:4 cement:sand mortar) across the panel cross-section:

- Shear area: ~85 mm × 1,000 mm = 85,000 mm²
- Conservative shear capacity: 85,000 × 0.3 = ~25 kN per panel
- With safety factor 3: ~**8 kN working load per panel**

With diagonal bracing (the designed configuration):

The diagonal bamboo strip (60 × 20 mm, Guadua angustifolia) in tension:

- Cross-section: 1,200 mm²
- Conservative tensile strength: 80 MPa (using ~50% of published 150–200 MPa)
- Diagonal tension capacity: 1,200 × 80 = ~96 kN
- Horizontal component (cos 69°): ~96 × 0.36 = ~34 kN per diagonal
- Two diagonals (one per side, one in tension): ~34 kN
- Wire ties and mortar interlock add additional resistance
- With safety factor 3: ~**12–15 kN working racking load per panel**

For context: A 3 m wide wall section (3 panels) provides ~36–45 kN of racking resistance. A single-story residential building in a moderate seismic zone (PGA 0.2g) with 30 kN of seismic mass above the wall needs roughly 6 kN of base shear resistance per linear meter of wall. Three panels per meter provide 4–5× this requirement.

These are very conservative numbers. The mortar matrix, wire tie grid, and)(profile all contribute additional resistance not accounted for above.

Out-of-Plane (Wind / Impact) Load

The panel resists pressure perpendicular to the wall face (wind suction/pressure, impact):

- The mortar arch created by the)(profile acts as a shallow vault between the rigid top and bottom T-bars
- Conservative estimate for uniform wind pressure, treating the panel as a simply supported slab:

Parameter	Value
Panel span (between top/bottom T-bars)	2.5 m
Panel width	1.0 m
Mortar thickness (minimum, at center)	~55 mm
Mortar flexural strength (conservative)	1.0 MPa
Estimated uniform pressure capacity	~0.7 kPa
With safety factor 2	~**0.35 kPa working load**

For context: 0.35 kPa ≈ wind speed ~25 m/s (90 km/h). For higher wind zones, the steel frame (not the panel) carries the wind load through the beam-to-column connections. The panel only needs to resist local pressure between its own frame members.

The bamboo strip grid and wire ties provide significant additional out-of-plane resistance not captured in the simple slab model — the panel is a composite, not a plain mortar slab.

Point Loads (Fixtures, Shelves, Cabinets)

Fixture type	Anchor method	Conservative capacity
Light items (pictures, clocks, towel hooks)	Masonry nail or screw into mortar	5–10 kg per anchor
Medium items (shelves, mirrors, small cabinets)	Plastic expansion anchor in mortar (6–8 mm)	15–25 kg per anchor
Heavy items (kitchen cabinets, water heater, TV mount)	Through-bolt to steel T-bar web	50–100 kg per anchor
Very heavy items (bookshelf, full pantry)	Multiple through-bolts to T-bar + distribution plate	200+ kg per mounting point

Tip: The T-bar web runs at a known position (center of wall thickness). Stud-finder or magnet locates it easily. Through-bolting to the web provides reliable, high-capacity anchorage comparable to steel stud framing.

Summary — Conservative Capacity in kg per Panel

For builders and non-engineers, here are the same numbers expressed in kilograms. These are **very conservative working loads** (already divided by safety factors of 2–3). Actual failure loads are 2–3× higher.

Load type	Direction	Conservative working capacity	What this means in practice
Racking (seismic/wind)	Horizontal, in the wall plane	**~1,200–1,500 kg per panel**	A horizontal force of 1.2–1.5 tonnes applied at the top of a single panel before it should be considered at its working limit. A typical single-story house needs ~600 kg of racking resistance per meter of wall — one panel provides 2× that.
Out-of-plane (wind pressure)	Perpendicular to wall face	**~90 kg spread over the panel face**	Equivalent to ~36 kg/m² uniform pressure. A person leaning hard against the wall (~80 kg on ~0.3 m²) is well within capacity.
Axial (vertical)	Downward, through the panel	**~4,000 kg per panel**	The two side T-bars carry 4 tonnes — 10–20× more than a typical roof + upper floor per meter of wall. Panels can carry a second story and roof without a steel frame.
Point load to mortar	Pull-out / shear at anchor	**~15–25 kg per anchor**	A shelf bracket with 2 anchors holds 30–50 kg safely.
Point load to T-bar	Through-bolt	**~50–100 kg per bolt**	A kitchen cabinet with 4 through-bolts holds 200–400 kg safely.

> ***Important:*** *These are estimates, not tested values. They will be validated through physical testing. Do not use for structural engineering without independent verification. When in doubt, attach heavy items to the steel building frame, not to the panel.*

Durability (80–100 Year Design Life)

Component	Degradation mechanism	Prevention	Expected life
Steel frame	Corrosion	Hot-dip galvanizing (85 μm+)	80–100 years (non-coastal)
Bamboo strips	Insect attack, fungal decay	Borate treatment + mortar encasement (alkaline, anaerobic)	80+ years

Component	Degradation mechanism	Prevention	Expected life
Mortar	Carbonation, weathering	Protected by lime plaster (sacrificial layer)	100+ years
Lime plaster	Surface erosion, micro-cracking	Self-healing (carbonation fills cracks), renewable lime wash	Indefinite with maintenance
Lime wash	UV degradation, rain erosion	Reapply every 5–10 years	5–10 years per coat
Electrical	Insulation degradation	PVC conduit allows cable replacement without opening wall	Cables: 30–50 years, replaceable
HDPE blocks	None significant	UV-protected (inside wall)	100+ years

Maintenance Schedule

Interval	Action	Estimated cost
Every 5–10 years	Lime wash renewal	$50–100 per building
Every 15–20 years	Inspect lime plaster, patch cracks	$100–300 per building
Every 30–40 years	Inspect electrical, replace cables if needed	$200–500 per building
At 40 years	Inspect steel frame at accessible points (spot re-galvanize if needed)	$200–400
Lifetime total		**$1,400–2,000**

Testing Needed

All load capacity values above are analytical estimates. They will be validated through physical testing as part of this project's development. The first panels will be built and tested before any building is occupied.

Planned testing program:

- [ ] **Racking shear tests** (ASTM E564 or equivalent) — panels with and without diagonals, to validate the 3–5× improvement estimate
- [ ] **Out-of-plane load tests** — uniform pressure and point impact, to validate the)(profile benefit
- [ ] **Fire resistance tests** (ASTM E119 or equivalent) — time to structural failure
- [ ] **Cyclic seismic tests** — full-scale shake table or quasi-static cyclic loading, to verify ductile failure mode
- [ ] **Connection tests** — panel-to-column bolt tab capacity, snap connector durability
- [ ] **Long-term weathering tests** — accelerated aging of mortar-encased bamboo samples
- [ ] **Acoustic transmission tests** (ASTM E90) — STC measurement
- [ ] **Point load pull-out tests** — masonry anchor and through-bolt capacity in mortar + T-bar

Test results will be published in this repository as they become available. If you have access to testing facilities and want to contribute, please get in touch — collaborative testing across different bamboo species and mortar mixes will make this system stronger for everyone.

Corrosion Prevention

Design Philosophy

The panel system uses multiple materials in intimate contact. Every material interface is a potential corrosion risk. This chapter documents the galvanic analysis of each interface and the prevention strategy.

Design target: 80–100 years without structural corrosion failure.

Galvanic Series (relevant metals)

From anodic (corrodes first) to cathodic (protected):

1. Zinc (galvanizing coating)
2. Aluminum (insect mesh)
3. Mild steel (frame, clamping strips)
4. Stainless steel (screws)

When two metals are in electrical contact in the presence of an electrolyte (moisture), the more anodic metal corrodes preferentially. This is the basis of galvanic corrosion.

Interface Analysis

1. Steel Frame ↔ Mortar

- **Contact:** Direct — mortar is poured onto and around the galvanized steel frame
- **Risk level:** LOW
- **Analysis:** The alkaline environment of cement mortar (pH 12–13) passivates the zinc coating, forming a stable zinc oxide layer. This is actually a *beneficial* combination — the mortar protects the zinc, and the zinc protects the steel. Same principle as rebar in reinforced concrete.
- **The pozzolanic admixture** gradually reduces mortar pH to 10–11 over decades. This remains well within the passivation range for zinc.
- **Prevention:** Hot-dip galvanizing per ISO 1461 (minimum 85 μm). No additional measures needed.

2. Stainless Steel Screws ↔ Galvanized Frame

- **Contact:** Direct — screws pass through the galvanized web
- **Risk level:** LOW-MEDIUM
- **Analysis:** Stainless steel is cathodic to zinc. In theory, this drives accelerated zinc corrosion at the contact point. In practice, the mortar encasement eliminates continuous moisture (the electrolyte), and the zinc coating at the screw hole is a sacrificial area — the surrounding zinc protects it cathodically.
- **Prevention:** Use A2 (304) stainless steel screws. The screw head is covered by the clamping strip and mortar. No moisture path.

3. Chicken Wire ↔ Steel Frame

- **Contact:** Indirect — chicken wire is stapled over the mortar face, separated from the frame by 30+ mm of mortar
- **Risk level:** NONE
- **Analysis:** Both are galvanized steel. No galvanic potential difference. Separated by mortar layer. No issue.

4. Aluminum Mesh ↔ Steel Frame

- **Contact:** Indirect — aluminum mesh sits between chicken wire and mortar, separated from frame by 30+ mm
- **Risk level:** NONE
- **Analysis:** Aluminum is anodic to steel (it would corrode first in direct contact). But there is no direct metal-to-metal contact — the mortar layer provides complete electrical isolation.
- **Prevention:** Ensure aluminum mesh does not contact the steel frame directly during installation. The mortar layer handles this naturally.

5. Aluminum Mesh ↔ Chicken Wire (Galvanized Steel)

- **Contact:** Direct — mesh layers touch during installation
- **Risk level:** LOW
- **Analysis:** Some galvanic potential exists. But both are embedded in mortar after installation, eliminating the moisture path. The aluminum mesh is a thin sacrificial layer — even if some corrosion occurs at contact points, the structural chicken wire remains intact.
- **Prevention:** No action needed. The mortar encasement is sufficient.

6. Clamping Strip ↔ Frame Web

- **Contact:** Direct — clamping strip is bolted against the web through bamboo strips
- **Risk level:** NONE
- **Analysis:** Both are hot-dip galvanized mild steel. Same material, same coating. No galvanic potential.

7. Steel Frame ↔ Bamboo Strips

- **Contact:** Direct — bamboo strips rest against the galvanized web
- **Risk level:** LOW
- **Analysis:** Bamboo is not a metal — no galvanic corrosion possible. The concern is moisture wicking through the bamboo creating a persistent damp zone on the steel. The mortar encasement eliminates this by sealing the interface. The borate treatment also reduces moisture absorption.

Environmental Factors

Coastal Environments (< 1 km from sea)

Salt-laden air accelerates zinc consumption dramatically. In coastal environments: - Increase galvanizing thickness to 120+ μm - Use 316L stainless steel screws (instead of 304) - Consider additional epoxy coating on frame before galvanizing - Expected frame life in coastal zone: 40–60 years (vs 80–100 inland)

High Rainfall (> 2,000 mm/year)

- Ensure adequate roof overhang (minimum 1.5 m) to keep rain off wall surfaces
- Lime plaster acts as sacrificial layer — renew every 10–15 years in high-rainfall zones
- Interior of wall (mortar-encased) is unaffected by rainfall

Tropical Humidity

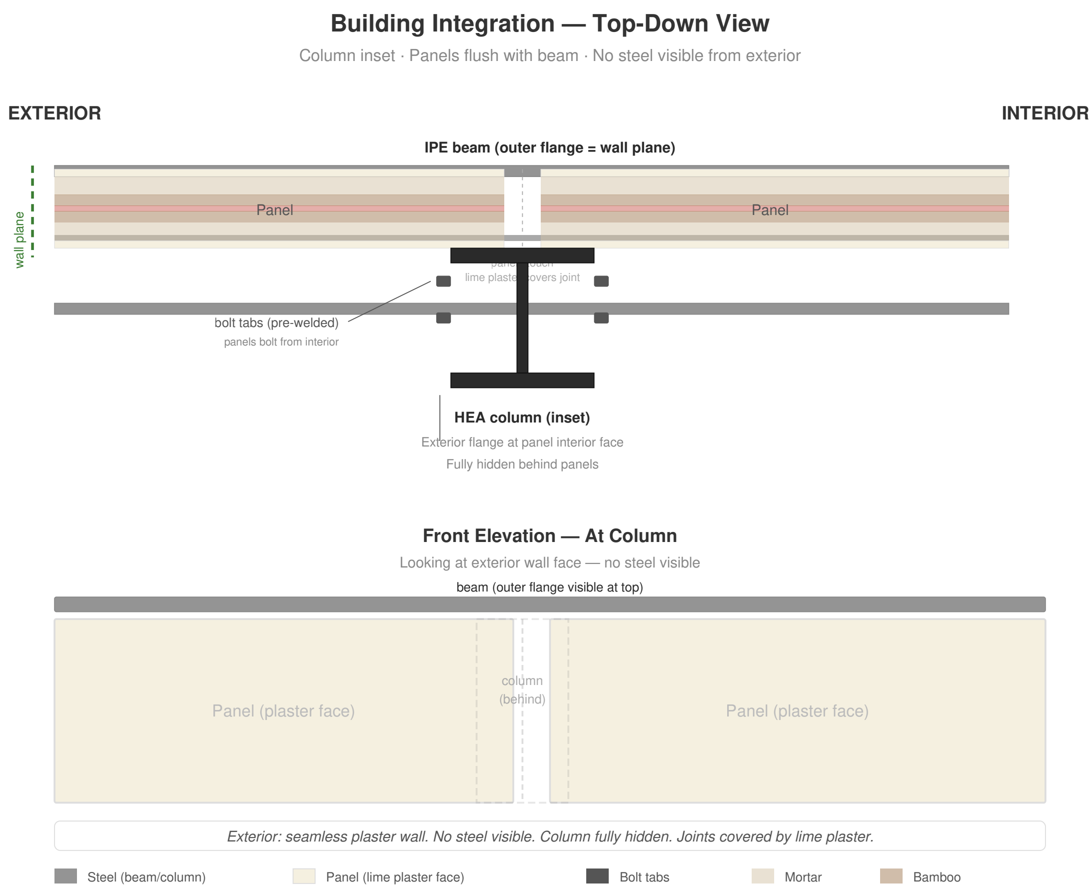
- Lime plaster's breathability is critical — it allows moisture to pass through rather than trapping it
- Do NOT seal the wall with paint or waterproof coating — this traps moisture and accelerates degradation
- The wall must breathe: lime wash only

Summary

Interface	Risk	Prevention	Maintenance
Steel frame in mortar	Low	Hot-dip galvanizing	None (inspect at 40 years)
SS screws in galv. frame	Low-Medium	A2 stainless, mortar encasement	None
Chicken wire to frame	None	Same material, mortar separation	None
Aluminum mesh to frame	None	Mortar isolation	None
Al mesh to chicken wire	Low	Mortar encasement	None
Clamping strip to frame	None	Same material	None
Steel to bamboo	Low	Borate treatment, mortar encasement	None

The system's corrosion protection relies on three layers: galvanizing (primary), mortar encasement (secondary), and lime plaster (tertiary/sacrificial). Failure of any single layer does not compromise the system.

Building Integration



Two Configurations

The panel system was designed to work **with a steel frame** but is structurally capable of standing on its own. Each panel can carry ~4,000 kg vertically — more than enough for a second floor and roof without steel columns.

Configuration	Best for	Structural role of panels
With steel frame	Multi-story, high seismic zones, commercial	Infill — steel carries primary loads, panels add bracing
Without steel frame	Single story + loft, moderate seismic, residential	Load-bearing — panels carry roof and upper floor directly, connected by ring beam

The sections below describe the steel-frame configuration in detail. For the frameless configuration, replace steel columns with a timber/bamboo ring beam at the top of the wall, and bolt panels to a concrete sill at the base and to each other at the edges.

Steel Frame Configuration

Frame Requirements

Element	Specification
Columns	HEA 150 or equivalent (sized per engineering calculation)
Beams	IPE 160–200 (spanning between columns)
Connections	Bolted on-site; bolt plates pre-welded in workshop
Column spacing	2–4 m (1 m increments = exact number of panels per bay)
Foundation	Reinforced concrete per local seismic code

Column-Panel Relationship

The key design detail: **columns are inset by the panel thickness** relative to the beams.

- The beam's exterior flange face = the wall plane
- Panels are flush with the beam face
- The column sits behind the panels, fully hidden
- From outside: continuous plaster wall with no visible steel

This is achieved by welding **bolt tabs** onto the column flanges at panel height. Panels bolt to these tabs from the interior side.

Panel Installation Sequence

1. **Steel frame erected** — columns, beams, roof structure complete
2. **Start from one corner** — bolt first panel to column tab
3. **Adjacent panel** — bolt to next tab, push edge against first panel (mortar to mortar)
4. **Snap-connect electrical** — 12V and 120V connectors click together at panel joint
5. **Continue around building** — panels fill each bay between columns
6. **Window/door openings** — skip panels, frame opening with steel lintels
7. **Test all circuits** — before finishing
8. **Lime plaster** — cover all panel faces and joints seamlessly
9. **Lime wash** — final finish

Panel-to-Column Detail (Top-Down View)

```
EXTERIOR                                INTERIOR

|← Lime plaster (3-5mm)
|← Mortar (15mm)
|← Chicken wire
|← Bamboo strips
|← T-bar web          ┌──────────┐
|← Mortar fill         │  COLUMN │  ← HEA 150
|← Bamboo strips       │ (hidden)│     (inset)
|← Chicken wire        │         │
|← Mortar (15mm)       └─────────┘
|← Lime plaster
```

```
Panel exterior face = Beam exterior face = Wall plane
```

Panel-to-Panel Joint

Adjacent panels touch **mortar face to mortar face**. The joint is:

1. Filled with mortar (if any gap exists)
2. Covered with chicken wire strip (bridging the joint)
3. Lime plastered over — completely invisible in finished wall

The electrical snap connectors are at the joint line. When panels are pushed together, the 2-pin (12V) and 3-pin (120V) connectors engage automatically.

Window and Door Openings

- Skip panels where openings are needed
- Frame the opening with steel lintels (welded to building columns/beams)
- Panel edges at openings remain exposed (steel frame edge visible) — covered with teak, wood, or steel trim
- Standard practice: openings at integer panel widths (1 m, 2 m, 3 m) for clean edges

Interior Partition Walls

Same panels, installed between interior columns or floor-to-ceiling brackets:

- Both sides visible and finished with lime plaster
- Full acoustic and fire separation
- Electrical circuits independent of exterior wall circuits
- Can be removed and reconfigured (panels unbolt from frame)

Roof Connection

- Panels stop at beam level (top of wall)
- Roof structure (purlins, rafters) sits on the beam
- The gap between panel top and roof underside is sealed with mortar or trim
- Roof overhang protects the wall from rain — minimum 1.5 m recommended in tropical climates

Floor Connection

- Panels sit on the ground floor slab or beam
- Base of panel is 30–50 mm above finished floor level (moisture protection)
- A galvanized steel base plate or hardwood sill bridges the gap
- Sealed with mortar and lime plaster

Electrical System Integration

When panels are installed sequentially, the snap connectors create continuous circuits:

- **12V lighting:** Runs through all panels. Connected to a central 12V transformer/power supply. Individual panel lights can be controlled by switches in Type S panels.
- **120V mains:** Runs through all panels. Connected to the building's main breaker panel. Circuit breakers sized per local electrical code.
- **Water (Type W panels):** Risers connect to building plumbing headers at floor level. Stub-outs are uncapped and connected to fixtures after installation.

Adaptability

The panel system accommodates different building configurations:

Configuration	Notes
Single story	Standard installation
Two story	Upper floor panels identical, bolt to upper beam frame
L-shaped or U-shaped plans	Corner panels: 90° junction with steel corner column
Curved walls	Not supported (panels are flat). Use faceted approximation with angled columns.
Mixed construction	Panels can fill part of a building, with other materials (glass, stone, timber) elsewhere

Contributing

Project Status

This project is **in active development**. The panel system has been designed and documented but is not yet built at full scale. The first physical build is planned in Colombia (Eje Cafetero region).

This means: - The documentation describes a design, not a tested product - Structural performance claims are based on engineering analysis, not laboratory testing - Your feedback and real-world experience are critical to improving the system

How to Contribute

Report Issues

Found an error, inconsistency, or design flaw? File an issue. Include: - Which document or diagram the issue is in - What you believe is wrong - Your suggested correction (if you have one)

Suggest Improvements

Have an idea that could make the system better? Open a discussion or issue with: - The problem you're solving - Your proposed solution - Any trade-offs or concerns

Share Build Experience

If you build with this system (even a single test panel): - Document your process with photos - Note any deviations from the documented method - Report what worked well and what didn't - Share material costs in your region

This real-world data is more valuable than any simulation.

Structural Testing

If you have access to testing facilities: - Racking shear tests (with and without diagonal bracing) - Out-of-plane load tests - Fire resistance tests - Cyclic seismic tests - Acoustic transmission tests

Test results — even informal ones — help build the evidence base for code compliance in different countries.

Bamboo Species Adaptation

The system was designed for *Guadua angustifolia* but should work with other structural bamboo species. If you adapt it for a different species: - Document the species, strip dimensions, and any process changes - Note any differences in behavior (flexibility, splitting, strength) - Share your results so others in your region can benefit

Translation

Help translate the documentation to more languages. Currently available: - English - German (Deutsch) - Spanish (Español)

To translate: fork the repository, create a new language folder (e.g., `fr/`, `pt/`, `id/`), translate all `.md` files, and submit a PR.

The SVG diagrams are designed to be language-neutral (numbered labels with legends), so they don't need translation.

Photography and Visualization

- Build process photos
- Finished panel close-ups
- Building integration photos
- 3D renderings
- Assembly animations

Visual documentation helps people understand the system faster than text alone.

Licensing

All contributions inherit the project's dual license: - Hardware designs: **CERN-OHL-S v2** (strongly reciprocal) - Documentation: **CC BY-SA 4.0** (share-alike)

By submitting a contribution, you agree to license it under these terms.

What this means for contributors: - Your work stays open — no one (including us) can close it - Others must credit you and share derivatives under the same license - Your improvements benefit every community that builds with this system

Code of Conduct

This project exists to help people build safe, dignified housing. Contributions should reflect that purpose:

- Be constructive — criticism of the design is welcome, but make it specific and actionable
- Be inclusive — the system is intended for communities with varying skill levels and resources
- Be honest — if something doesn't work, say so. False claims of performance help no one.
- Share freely — don't submit contributions that contain third-party intellectual property restrictions

Contact

- **Designer:** Jan Reuteler (janreuteler@icloud.com)
- **Repository issues:** The best way to contribute technically
- **General questions:** Email the designer directly

BaharequeModular is a small project with big ambitions. Every contribution — from a typo fix to a structural test report — moves it forward.

Cost Estimate

> ***Disclaimer:*** *These are material cost estimates only, based on 2025 prices in Colombia (Eje Cafetero region). Labor is not included — the system is designed for community production where labor is contributed, not purchased. Prices vary significantly by region, currency, and supply chain. Use these as a starting point, not a budget.*

Cost Per Panel (materials only)

Material	Specification	Qty per panel	Unit cost (USD)	Total (USD)
T-bar steel	30×30×3 mm, galvanized	7.0 m	$2.50/m	$17.50
Galvanizing	Hot-dip, batch process	1 frame	$3.00	$3.00
Clamping strips	2 mm flat steel, galv.	4.0 m	$1.00/m	$4.00
SS screws	Self-tapping, A2	40 pcs	$0.08	$3.20
HDPE blocks	Recycled, 30×30 mm	2.0 m	$1.50/m	$3.00
Bamboo strips (vertical)	Borate-treated guadua	~135 m total	$0.05/m	$6.75
Bamboo strip (diagonal)	60×20 mm, treated	5.4 m	$0.30/m	$1.60
Wire ties	Galvanized	5 m	$0.10/m	$0.50
PVC conduit	16 mm electrical	5 m	$0.30/m	$1.50
12V cable + sockets	2-core + 6× E10	1 set	$3.00	$3.00
120V cable + connectors	3-core + snap connectors	1 set	$4.00	$4.00
Chicken wire mesh	Galvanized hex, 25 mm	5.5 m²	$0.80/m²	$4.40
Cement	Portland Type I	~25 kg	$0.15/kg	$3.75
Sand	Clean river sand	~100 kg	$0.02/kg	$2.00
PP fiber	Chopped 6–12 mm	200 g	$5.00/kg	$1.00
Pozzolanic admixture	Volcanic ash or RHA	2 kg	$0.50/kg	$1.00
Panel subtotal				**$60–65**

On-site finishing (per panel area)

Material	Qty per panel	Cost (USD)
Lime plaster (both faces)	~15 kg lime + 40 kg sand	$4.00
Lime wash (both faces)	~2 kg lime	$0.50
Finishing subtotal		**$4.50**

Total material cost per panel: ~$65–70 USD

Cost per m² of finished wall

- Panel area: 1.0 m × 2.5 m = 2.5 m²
- Material cost per m²: ~**$26–28 USD/m²**

Comparison with Conventional Construction

Wall system	Material cost per m² (USD)	Lifespan	Notes
This panel system	**$26–28**	**80–100 years**	Includes structure, electrical, finish
Concrete block (15 cm)	$25–35	40–60 years	Requires rebar, grout, plaster, paint, separate electrical
Fired brick	$30–45	60–80 years	Requires mortar, plaster, paint, separate electrical
Drywall on steel studs	$20–30	20–30 years	No thermal mass, not load-bearing, requires paint, separate electrical
Conventional bahareque (in-situ)	$15–25	15–30 years	No electrical, finish quality variable, short lifespan without engineering
Adobe block	$10–20	20–40 years	Poor seismic performance, no electrical, requires thick walls

Key insight: The panel system costs roughly the same as concrete block per m² — but includes integrated electrical, has 2× the lifespan, provides better thermal and acoustic performance, and produces a superior finish. When you factor in the cost of wiring a concrete block house separately, the panel system is cheaper.

Cost Per House (rough estimate)

A simple single-story house with loft (50 m² footprint, ~70 m² livable including loft):

Component	Quantity	Cost (USD)
Wall panels	~40 panels (100 m² of wall)	$2,600–2,800
Foundation (concrete, rebar)	~5 m³ concrete	$800–1,200
Ring beam (timber/bamboo)	~30 m	$200–400
Roof (terracotta tiles + purlins)	~65 m²	$1,200–1,800
Floor (concrete slab or earthen)	~50 m²	$400–800
Windows + doors	~8 openings	$600–1,200
Electrical panel + transformer	1 set	$200–400
Plumbing (basic)	Kitchen + bathroom	$300–600
Total materials		**$6,300–9,200**

This is materials only. In a community-build model, labor is contributed. In a contractor model, add 40–60% for labor.

For context: A comparable concrete block house in rural Colombia costs $12,000–20,000 (materials + labor) and lasts 40–60 years. This system achieves similar or better quality at roughly half the cost with twice the lifespan.

Cost Reduction Strategies

Strategy	Savings	Notes
Harvest own bamboo	30–40% of bamboo cost	Requires 1+ hectares of guadua stand, 4–5 year maturity
Recycle HDPE from waste	100% of HDPE cost	Cutting boards, pipe offcuts, industrial waste
Use volcanic ash (free)	100% of pozzolan cost	Available in Andean regions, collect and sieve

Strategy	Savings	Notes
Use star grass fiber instead of PP	100% of fiber cost	Invasive grass, free, circular use
Batch galvanizing (50+ frames)	15–20% of galv. cost	Volume discount from galvanizing plant
Community sand quarry	50–80% of sand cost	If river sand access is available locally
Make own lime from limestone	50–70% of lime cost	Requires kiln (one-time build), local limestone

Regional Price Variations

Material costs vary dramatically by region. The estimates above are for Colombia. Rough multipliers:

Region	Multiplier	Notes
Colombia (Eje Cafetero)	1.0× (baseline)	Guadua abundant, volcanic ash available
Ecuador, Peru	0.8–1.1×	Similar materials, caña guadua available
Central America	1.0–1.3×	Bamboo available, steel may cost more
Philippines, Indonesia	0.7–1.0×	Bamboo abundant and cheap, labor very low
India	0.6–0.9×	Bamboo and labor very low, steel affordable
Sub-Saharan Africa	0.8–1.2×	Bamboo species vary, cement may be expensive
Southern Europe	2.0–3.0×	Materials available but more expensive

These multipliers are rough guides. The single biggest variable is bamboo cost — if you grow your own, it's nearly free.

Environmental Impact

Carbon Footprint Per Panel

Material	Qty per panel	Embodied CO_2 (kg)	Source of emission
Portland cement	25 kg	20.0	Calcination + kiln fuel (~0.8 kg CO_2/kg cement)
Steel T-bar + clamping strips	~8 kg steel	14.4	Steelmaking (~1.8 kg CO_2/kg steel)
Galvanizing (zinc)	~0.3 kg zinc	1.2	Zinc smelting (~4 kg CO_2/kg zinc)
Sand	100 kg	0.5	Extraction + transport
Chicken wire	~1.5 kg	2.7	Steel wire production
PVC conduit	~0.3 kg	0.6	PVC manufacturing (~2 kg CO_2/kg)
Cables + connectors	~0.5 kg	1.5	Copper + plastic
HDPE blocks	~0.5 kg	0.5	If recycled: ~1 kg CO_2/kg (vs 3 for virgin)
Subtotal emissions		**~41 kg CO_2**	
Bamboo (carbon sequestered during growth)	~10 kg dry weight	**–15 to –20 kg CO_2**	Bamboo sequesters ~1.5–2 kg CO_2 per kg dry mass
Lime plaster (reabsorbs CO_2 during carbonation)	~15 kg lime	**–5 to –8 kg CO_2**	Lime carbonation recaptures ~30–50% of calcination CO_2
Net CO_2 per panel		**~15–25 kg CO_2**	
Net CO_2 per m² of wall		**~6–10 kg CO_2/m²**	

Comparison with Conventional Wall Systems

Wall system	kg CO_2 per m²	Ratio vs this system
This panel system	**6–10**	**1× (baseline)**
Concrete block (15 cm, plastered)	40–60	5–7×
Fired clay brick (plastered)	50–80	6–10×
Reinforced concrete wall (15 cm)	70–100	8–12×
Drywall on steel studs	15–25	2–3×
Conventional bahareque (mud plaster)	3–8	0.5–1×
Adobe block	5–10	0.7–1×

The panel system produces **5–10× less CO_2 than concrete block or brick** — the two most common wall materials in the developing world. It is comparable to traditional earth construction but with vastly superior structural performance and lifespan.

Carbon Per Year of Service

When lifespan is factored in, the comparison becomes even more dramatic:

Wall system	kg CO_2/m^2	Lifespan (years)	kg CO_2/m^2/year
This panel system	8	90	**0.09**
Concrete block	50	50	1.00
Fired brick	65	70	0.93
Drywall on steel studs	20	25	0.80
Conventional bahareque	5	20	0.25

Over its lifetime, this panel system produces ~10× less carbon per year of service than concrete block.

Path to Carbon-Neutral or Carbon-Negative

The panel system is already low-carbon. With the following substitutions, it approaches carbon-neutral:

Substitution	CO_2 saved per panel	Feasibility
Replace 50% cement with pozzolan (volcanic ash, RHA)	−10 kg	Available in volcanic and rice-growing regions
Use recycled steel (electric arc furnace)	−7 kg	Depends on local steel supply chain
Replace PP fiber with star grass	−0.5 kg	Available anywhere the grass grows
Increase bamboo content (thicker strips, more diagonals)	−3 to −5 kg additional sequestration	Material is free/cheap
Use bio-lime (wood-fired kiln with coppiced fuel)	−3 kg	Traditional lime production method
Total potential reduction	**−20 to −25 kg**	

With these substitutions, net CO_2 per panel drops to **~0–5 kg** — effectively carbon-neutral. If bamboo content is increased further or the bamboo stand is managed as a carbon sink, the system becomes **carbon-negative**: the wall absorbs more CO_2 during its life than was emitted to build it.

Water Use

- **Mortar mixing:** ~15–20 liters per panel (curing water reused where possible)
- **Curing:** ~50 liters over 7 days (misting/burlap method)
- **Lime plaster mixing:** ~10 liters per panel
- **Total:** ~75–80 liters per panel, or ~30 liters per m² of wall

For comparison: concrete block production uses ~40–50 liters per m² of wall. Fired brick production uses ~100+ liters per m² (including kiln cooling).

Waste

- **Steel offcuts:** Recycled (scrap value)
- **Bamboo offcuts:** Composted or used as garden stakes/mulch
- **Mortar waste:** Minimal — vibration table pour is precise. Any excess is reused in the next panel.
- **Sand bed:** Reused indefinitely (sand is not consumed, just compressed and loosened)
- **HDPE offcuts:** Recycled or reused for smaller blocks
- **Packaging waste:** Minimal — most materials arrive in bulk

The system produces **near-zero construction waste**. This is a direct consequence of workshop prefabrication — every cut is planned, every material is measured.

End of Life (after 80–100 years)

When a panel eventually reaches end of life:

Material	Disposal	Environmental impact
Steel frame	Recycle (100% recyclable)	Positive — scrap steel has value
Mortar + bamboo	Crush and use as aggregate or fill	Inert — mortar is calcium/silica/sand, bamboo is organic
Lime plaster	Returns to calcium carbonate (limestone)	Carbon-neutral — CO_2 absorbed during carbonation is re-released, net zero
HDPE blocks	Recycle	Recyclable
Copper cables	Recycle (high scrap value)	Positive
PVC conduit	Recycle where facilities exist	Recyclable in principle

No hazardous waste. No asbestos, no lead paint, no synthetic insulation foam, no pressure-treated wood chemicals. The panel is made of steel, bamboo, sand, cement, and lime — it returns to the earth or the recycling stream cleanly.

Transport & Handling

Panel Specifications for Handling

Property	Value
Dimensions	1.0 m × 2.5 m × 85 mm
Weight	~155 kg
Center of gravity	Center (symmetric construction)
Fragile areas	Corners, edges (mortar can chip if impacted)
Minimum cure before handling	7 days
Minimum cure before installation	14 days recommended

Carrying by Hand

A panel at 155 kg requires **3–4 people** using a carrying jig:

Simple Carrying Jig

Two bamboo poles (3 m long, 8–10 cm diameter) threaded through loops of heavy webbing strap wrapped around the panel at 1/4 and 3/4 height:

1. Lay panel flat
2. Wrap two webbing straps (50 mm wide, rated 500+ kg) around the panel at ~60 cm and ~190 cm from the bottom
3. Thread a bamboo pole through both strap loops on each side
4. Two people per pole, lift from the ends
5. Panel hangs vertically between the poles

Weight per person: ~40 kg — manageable for short distances (workshop to truck, truck to site).

For longer carries (>50 m), add a third pole with a center strap for 6 carriers (~26 kg each).

Lifting and Tilting

- **Never lift a panel by one edge** — the mortar can crack under its own weight if cantilevered
- **Always lift from two points** symmetrically, or from the bottom while supported at the back
- **Tilting from flat to vertical:** Two people hold the top edge, two people lift the bottom. Rotate slowly. Support the bottom on a padded block during rotation.

Workshop Storage

After curing, panels are stored **vertically (on edge)** in an A-frame rack:

- Lean panels against both sides of an angled frame (like an open book)
- 10–15 panels per side of the rack
- Separate panels with small wooden spacers to allow air circulation
- Cover with tarp or roof to protect from direct rain (curing mortar can be washed out if hit by heavy rain before full cure)
- Keep off the ground — place bottom edge on gravel or wooden rail

Space required: ~3 m × 1 m floor area per 20–30 panels stored.

Truck Transport

Loading

1. Stand panels vertically in the truck bed, leaning against one side wall
2. Place wooden spacers (offcuts, bamboo, or foam) between each panel
3. Strap panels to the truck side wall using ratchet straps at 1/3 and 2/3 height
4. Never stack panels flat on top of each other — the bottom panel will crack under the weight of the stack
5. Maximum ~15–20 panels per standard pickup truck (depending on truck capacity and road conditions)

Road Considerations

- **Paved roads:** No special precautions beyond strapping
- **Unpaved/rough roads:** Reduce speed. Add foam or cardboard padding between panels. Strap tightly — panels can shift and chip edges on bumps.
- **Steep grades:** Ensure panels lean into the hill (not away from it) to prevent toppling
- **Distance:** No limit. Panels are fully cured and self-contained. They can travel any distance.

Unloading

1. Remove straps from the top down
2. Two people per panel — one at top edge, one supporting bottom
3. Walk each panel to the A-frame storage rack at the site
4. Never drop or slide panels — mortar edges chip easily

Crane / Hoist (for upper floors)

For two-story buildings, panels must be lifted to the second floor:

- **Simple gin pole:** A bamboo or steel pole with a rope and pulley. Two people pull the rope, one guides the panel. Works for low volumes.
- **Chain hoist on beam:** Attach a 500 kg chain hoist to the building's steel beam. Clip to the panel's carrying strap. One person operates, one guides. Fast and controlled.
- **Forklift (if available):** Fork under the panel (protected with cardboard). Lift and place directly into position.

Do not throw, roll, or drag panels. The mortar is strong in compression but can crack on point impact.

Damage and Repair

Damage type	Cause	Repair
Corner chip (small)	Impact during transport	Fill with mortar, plaster over. Cosmetic only.
Edge crack (surface)	Bumping during handling	Fill with lime plaster. Does not affect structure.
Through-crack (mortar)	Dropped or severe impact	Panel may be compromised. Assess: if steel frame is intact and crack is <2 mm, mortar-fill and monitor. If crack is >5 mm or frame is bent, reject panel.
		Replace connector (soldered or crimped).

Damage type	Cause	Repair
Broken snap connector	Snagged during transport	

Rejection rate: With reasonable care, expect <5% of panels to suffer any damage during transport and handling, and <1% to require rejection. Rejected panels can be stripped of electrical components and recycled as fill material.

Checklist Before Transport

- [ ] Panel cured minimum 7 days (14 recommended)
- [ ] Panel marked with type (P/O/S/W) and production date
- [ ] Electrical circuits tested (12V + 120V continuity check)
- [ ] No visible cracks >1 mm
- [ ] Corners intact
- [ ] Carrying straps or jig prepared
- [ ] Truck bed clean, spacers ready, straps available
- [ ] Route checked for road conditions (bridges, clearance, grade)

Climate Adaptation

The panel system was developed for the Colombian Andes (~2,000 m altitude, ~2,500 mm annual rainfall, 15–25°C year-round). This chapter documents adaptations for other climates.

Climate Zones

Tropical Humid (0–1,000 m, >2,000 mm rain, 25–35°C)

Examples: Pacific coast Colombia, Amazon basin, Philippines, Indonesia, West Africa

Factor	Adaptation
Rain	Minimum 2.0 m roof overhang on all sides. Rain splashing off ground erodes lime plaster at the base — raise panel bottom 30–50 cm above grade on a concrete or stone plinth.
Humidity	Lime plaster is critical — it breathes. **Never** seal the wall with paint, acrylic render, or waterproof coating. Trapped moisture destroys both bamboo and steel.
Insects	Borate treatment of bamboo is non-negotiable. Increase borate concentration to 8–10% (vs standard 5%). Mortar encasement provides secondary protection. Termite barriers at foundation level.
Fungal growth	Lime wash is naturally antifungal. Reapply every 3–5 years in humid lowlands (vs 5–10 in highlands). Add copper sulfate at 1–2% to lime wash for additional fungal resistance.
Heat	The 85 mm mortar core provides excellent thermal mass — absorbs heat during the day, releases at night. Combine with cross-ventilation (window openings on opposite walls).
Mortar mix	Standard 1:4 cement:sand works well. Increase pozzolan content to 15% (reduces heat of hydration and improves long-term durability in wet conditions).
Lime plaster	Use hydraulic lime (NHL 3.5 or NHL 5) rather than air lime — it sets faster and resists rain better during application.

Tropical Highland (1,000–2,500 m, 1,500–2,500 mm rain, 12–25°C)

Examples: Colombian Eje Cafetero, Ecuadorian Sierra, Ethiopian highlands, Kenyan highlands

This is the **design climate** — no significant adaptations needed.

Factor	Notes
Rain	1.5 m overhang sufficient. 2.0 m for exposed facades.
Temperature	Thermal mass smooths day-night swings. No insulation needed.
Humidity	Moderate — lime plaster breathes adequately.
Insects	Standard borate treatment (5%).
Lime wash	Every 5–10 years.
Mortar mix	Standard 1:4 + 10% pozzolan.

Tropical Dry / Semi-Arid (0–1,500 m, <800 mm rain, 25–40°C)

Examples: Caribbean coast Colombia, Northeast Brazil, Sahel, Northern India, parts of Mexico

Factor	Adaptation
Rain	Less concern. 1.0 m overhang sufficient.

Factor	Adaptation
Heat	Thermal mass is excellent here — walls absorb daytime heat and release at night. Light-colored lime wash reflects solar radiation.
UV exposure	Intense sun degrades lime wash faster. Reapply every 3–5 years. Consider deeper overhangs on sun-facing walls.
Mortar curing	Critical — dry air causes rapid moisture loss during curing. Cover panels with wet burlap and plastic for 7+ days. Cure in shade, never in direct sun.
Dust	Fine dust accumulates in lime plaster texture. Periodic wash with water keeps surfaces clean.
Bamboo sourcing	Guadua may not grow locally. Use alternative species (see Materials) or import treated bamboo.

Subtropical (25–35° latitude, seasonal temperature variation, 800–1,500 mm rain)

Examples: Southern Brazil, Northern Argentina, Southern China, Southeast USA, Northern India, South Africa

Factor	Adaptation
Winter cold	The mortar core provides thermal mass but not insulation. For climates with sustained cold (<5°C for weeks), add external insulation: 30–50 mm of wood fiber board or cork outside the lime plaster, with a breathable render over it. **Never** insulate on the interior — this traps moisture in the wall.
Freeze-thaw	If temperatures drop below 0°C, use hydraulic lime (NHL 5) for plaster. Air lime can suffer freeze-thaw damage before full carbonation. Mortar core (inside the wall) is protected by plaster and thermal mass — not at risk.
Seasonal rain	Standard overhang. Ensure base of wall is raised above grade and protected from splash.
Bamboo species	Guadua angustifolia does not grow above ~30° latitude. Use *Phyllostachys* species (Moso, P. bambusoides) — smaller diameter, wider strips needed. See Materials.

Coastal (any latitude, <2 km from sea)

Examples: Caribbean islands, Pacific coast, Mediterranean coast, Southeast Asian coast

Factor	Adaptation
Salt air	Dramatically accelerates zinc corrosion. Increase galvanizing to 120+ μm. Use 316L stainless steel screws (not 304). Consider additional epoxy primer on frames before galvanizing.
Wind	High wind zones require the steel frame configuration (not frameless). Design building frame for local wind code. Panel racking resistance contributes to overall building stiffness.
Humidity + salt	Lime wash every 3–5 years. Inspect plaster annually for salt crystallization damage (efflorescence). Brush off salt deposits before re-washing.
Corrosion	Expected frame life in coastal zone: 40–60 years (vs 80–100 inland). At 40 years, inspect and re-galvanize exposed steel if needed. See Corrosion Prevention.
Storm surge / flooding	Raise floor level well above flood line. Panels survive brief immersion (mortar is water-resistant) but prolonged submersion degrades embedded bamboo. Design for elevated construction.

Mortar Mix Adjustments by Climate

Climate	Cement:Sand	Pozzolan	PP Fiber	W/C Ratio	Curing
Tropical humid	1:4	15%	Standard	0.45	7 days, covered
Tropical highland (default)	1:4	10%	Standard	0.45–0.50	7 days, moist

Climate	Cement:Sand	Pozzolan	PP Fiber	W/C Ratio	Curing
Tropical dry	1:4	10%	Standard	0.50	10 days, wet burlap + plastic, shade
Subtropical (cold winters)	1:3.5	10%	Standard	0.45	7 days; do NOT pour if temp <5°C
Coastal	1:4	15%	Standard	0.45	7 days; use fresh water only, no sea sand

Lime Plaster Adjustments

Climate	Lime type	Mix	Fiber	Maintenance interval
Tropical humid	NHL 3.5 or NHL 5	1:2.5	Grass fiber	Every 3–5 years wash
Tropical highland	NHL 3.5 or air lime	1:3	Optional	Every 5–10 years wash
Tropical dry	Air lime or NHL 3.5	1:3	Grass fiber (crack prevention)	Every 3–5 years wash (UV)
Subtropical	NHL 5	1:2.5	Grass fiber	Every 5–7 years wash
Coastal	NHL 5	1:2.5	Grass fiber	Every 3–5 years (salt)

Roof Overhang Summary

The single most important climate adaptation — protecting the wall from rain:

Climate	Minimum overhang	Notes
Tropical humid	2.0 m	Non-negotiable. Heavy rain + wind-driven spray.
Tropical highland	1.5 m	Standard. 2.0 m on exposed facades.
Tropical dry	1.0 m	Sun protection more important than rain.
Subtropical	1.5 m	Rain protection + summer shade.
Coastal	2.0 m	Rain + wind-driven salt spray.

Bamboo Species by Climate Zone

Climate zone	Recommended species	Notes
Tropical Americas (0–2,000 m)	*Guadua angustifolia*	Strongest. Abundant in Colombia, Ecuador, Central America.
Tropical SE Asia	*Dendrocalamus asper*, *Bambusa bambos*	Large diameter. Direct substitute for guadua.
Tropical South Asia	*Bambusa bambos*, *Dendrocalamus strictus*	Widely available. D. strictus is denser but smaller.
Tropical Africa	*Bambusa vulgaris*, *Oxytenanthera abyssinica*	B. vulgaris is introduced but common. O. abyssinica is native.
Subtropical / Temperate	*Phyllostachys edulis* (Moso), *P. bambusoides*	Smaller diameter — use wider, thinner strips.
Temperate cool	Limited bamboo options	Consider importing treated bamboo strips, or substitute with other natural fibers/laths.

Bamboo Treatment Protocol

Untreated bamboo is eaten by powder-post beetles and degraded by fungi within 1–5 years in tropical climates. Proper treatment extends bamboo life to **decades** — and when encased in mortar inside a panel, to **80+ years**.

This chapter covers borate treatment — the safest, cheapest, and most effective method for bamboo used in construction.

Why Borate

Property	Borate	CCA (chromated copper arsenate)	Creosote
Toxicity to humans	Very low (borax is used in soap)	High (arsenic, chromium)	High (carcinogenic)
Insect protection	Excellent	Excellent	Excellent
Fungal protection	Good	Excellent	Excellent
Cost	Very low	Moderate	Moderate
Environmental	Safe, water-soluble	Toxic waste	Toxic waste
Suitability for bamboo	Excellent (penetrates well)	Poor (bamboo resists oil-based)	Poor
Compatibility with mortar	Excellent (no chemical reaction)	Unknown	Incompatible

Borate is the clear choice for bamboo in construction. It is non-toxic, cheap, effective, and compatible with the mortar encasement that provides secondary long-term protection.

Borate Solution Preparation

Materials

Material	Quantity per 100 liters	Notes
Borax (sodium tetraborate, $Na_2B_4O_7 \cdot 10H_2O$)	3 kg	Available at agricultural supply stores, pharmacies, or industrial chemical suppliers
Boric acid (H_3BO_3)	2 kg	Same sources as borax
Water	100 liters	Clean, potable. Warm water (40–50°C) dissolves faster.

This produces a ~**5% borate solution** (by weight of dissolved solids). For high-insect-pressure environments (tropical humid lowlands), increase to 4 kg borax + 3 kg boric acid per 100 L for an ~**7% solution**.

Mixing

1. Heat 20 liters of water to 50–60°C (hot but not boiling)
2. Add borax, stir until fully dissolved (~5 minutes)
3. Add boric acid, stir until dissolved (~5 minutes)
4. Add remaining 80 liters of room-temperature water
5. Stir well. Solution should be clear or slightly cloudy.
6. Solution is ready to use immediately.

Shelf life: Indefinite if stored in a closed container. Borate does not degrade.

Treatment Methods

Method 1: Tank Soaking (simplest, recommended for strips)

Best for: split bamboo strips (the form used in this panel system).

1. **Prepare strips:** Split and clean bamboo. Remove any remaining inner membrane (soft tissue). Strips should be ~20 mm wide, 2,500 mm long.
2. **Build or find a tank:** A simple trough from concrete blocks and plastic sheet, a PVC pipe cut lengthwise, or a purchased livestock water trough. Must hold strips fully submerged.
3. **Submerge strips completely** in borate solution. Weigh down with stones or a bamboo lattice if they float.
4. **Soak for 5–7 days** at room temperature. Borate diffuses into the bamboo tissue. Thinner strips (like the 20 mm strips used here) absorb faster than whole culms.
5. **Remove and dry.** Stack strips on a rack in shade with air circulation. Dry for 3–7 days until moisture content drops below 15%.
6. **Visual check:** Treated bamboo often shows a slight greenish or bluish tint from the borate. This is normal and confirms penetration.

Solution reuse: The same tank of solution can be used for multiple batches. Top up with fresh solution as the level drops (bamboo absorbs some). Test concentration with a borate test kit if available, or replace every 5–6 batches.

Capacity: A 3 m × 0.5 m × 0.3 m trough holds ~450 liters and treats ~200 strips per batch — enough for ~4 panels.

Method 2: Modified Boucherie (for whole culms before splitting)

Best for: treating whole guadua culms before splitting into strips. Faster penetration through the vascular system.

1. **Harvest fresh bamboo** (within 24 hours of cutting — sap flow must still be active)
2. **Cut culm to length.** Pierce all internal diaphragms (nodes) with a steel rod.
3. **Attach a rubber cap or hose** to the base end of the culm, connected to a raised tank of borate solution (gravity-fed, 2–3 m elevation difference).
4. **Solution flows through the culm** via the vascular bundles, pushed by gravity. The borate-rich solution displaces the natural sap and exits from the top end as light-colored drips.
5. **Treatment time:** 3–5 days for a 6 m culm. Complete when the dripping liquid from the top tests positive for borate (use turmeric indicator — turns red in borate).
6. **Remove, dry, and split** into strips after treatment.

Advantages: More uniform penetration than soaking. Works for thick-walled culms where soaking would take weeks. Preserves sap-path moisture channels that aid borate distribution.

Disadvantages: Requires fresh-cut bamboo (not dried). Requires elevated tank and hose setup. Not practical for already-split strips.

Method 3: Spray/Brush (emergency only, not recommended)

Surface application of borate solution by spraying or brushing. Penetration is limited to 1–3 mm. **Not adequate for structural bamboo.** Use only as a temporary measure if tank soaking is not available, and re-treat properly at the first opportunity.

Quality Control

Penetration Test

After treatment and drying, test a sample strip:

1. Cut a cross-section from the middle of a treated strip

2. Apply **turmeric solution** (dissolve 1 teaspoon turmeric powder in 50 ml rubbing alcohol)
3. Brush onto the cut cross-section
4. Borate-treated wood turns **red/dark orange** on contact with turmeric
5. Untreated wood remains **yellow**

The entire cross-section should turn red — confirming full penetration. If the center remains yellow, increase soak time for the next batch.

Retention Test (advanced)

For formal quality control, send treated samples to a lab for borate retention analysis (target: >4 kg BAE/m^3 — boric acid equivalent per cubic meter of bamboo). This level is sufficient for all wood-boring insects and most fungal species.

Safety

- **Borate is low-toxicity.** Borax is used in commercial cleaning products. Boric acid is used in eye wash solutions. Neither is classified as hazardous at the concentrations used.
- **Wear gloves** when handling concentrated solution (mild skin irritant with prolonged contact).
- **Do not drink the solution.** Borate is mildly toxic if ingested in quantity (comparable to table salt). Keep away from children and animals.
- **Do not discharge used solution into waterways.** Borate can affect aquatic organisms at high concentrations. Dispose of used solution by applying it to the ground away from water sources (the boron is a plant micronutrient at low concentrations).

Treatment Summary for This Panel System

Bamboo component	Treatment method	Soak time	Concentration
Vertical strips (20 mm × 2,500 mm)	Tank soak	5–7 days	5% (standard) or 7% (humid lowland)
Diagonal strips (60 × 20 mm × 2,690 mm)	Tank soak	7–10 days (thicker)	5–7%
Whole culms (before splitting)	Modified Boucherie	3–5 days	5%

After treatment and drying, bamboo is ready for panel assembly. The borate provides primary insect/fungal protection. The mortar encasement provides secondary protection — creating an alkaline, anaerobic environment hostile to biological degradation.

Combined protection lifespan: 80+ years.

Acoustic Upgrade — Noise-Facing Walls

> *This chapter was inspired by a suggestion from Kim Hansson — thank you for pushing the design to address real-world urban conditions.*

The standard panel (STC 40–45) is adequate for quiet residential areas. For buildings on busy streets, near highways, or adjacent to commercial/industrial zones, the noise-facing walls can be upgraded individually. There is no need to upgrade every wall — most buildings only have one or two sides exposed to significant noise. The quiet-facing walls remain standard panels.

Noise Exposure Guide

Situation	Recommended treatment	Target STC
Quiet residential street	Standard panel — no upgrade needed	40–45
Moderate street traffic	Dense mortar + clay damping layer	48–52
Busy urban road, bus routes	Double panel with fiber cavity	55–62
Highway, industrial, nightlife district	Double panel with fiber cavity + clay layer	58–65

Apply the upgrade only where needed. A typical house on a busy street has one or two noise-facing walls (10–15 m²). The remaining walls face a garden, courtyard, or quiet side and use standard panels. This keeps the cost increase proportional to the actual noise problem.

Option 1: Denser Mortar Mix (cheapest, +3–5 STC)

Replace standard river sand with heavier aggregates in the mortar mix. Same production process, same frame, same vibration table pour. The panel is simply heavier.

Aggregate	Density increase	Cost vs standard sand	Availability
Crushed basalt or granite fines	+15–20%	Similar	Universal — any quarry
Iron slag (*escoria*)	+25–30%	Often free	Near steel mills, foundries
Magnetite sand (*arena negra*)	+40%	Low	Volcanic river beds (Andes, Central America)

- **Panel weight:** ~175–190 kg (vs ~155 kg standard)
- **Extra cost:** ~$1–2 per panel
- **STC:** ~43–48
- **Production change:** None — just substitute the aggregate. Mix ratio remains 1:4 cement:aggregate.

Best for: Side walls that need a modest improvement without any change to wall thickness or construction process.

Option 2: Clay Damping Layer (+5–8 STC)

Add a 10–15 mm layer of **clay-straw plaster** on the interior face of the noise-facing wall, applied on-site between the mortar surface and the lime plaster finish.

Why Clay Works

Clay has superior acoustic damping compared to cement mortar. When sound waves hit a clay layer, more energy is converted to heat (internal friction in the clay particles) rather than being transmitted through the wall. Chopped straw fibers within the clay add additional internal friction.

This is the most traditional bahareque technique — clay plaster (*pañete de tierra*) has been used on bamboo walls for centuries. It works acoustically for the same reason it works thermally: clay is a natural energy absorber.

Application

1. Prepare a clay-straw mix: local clay + chopped dried grass or straw (5–10 cm lengths), mixed to a thick paste
2. Apply 10–15 mm to the interior face of the noise-facing wall, over the mortar surface
3. Allow to dry (3–7 days depending on humidity)
4. Apply lime plaster (3–5 mm) over the clay layer as the finish coat
5. Lime wash as usual

Specifications

- **Total wall thickness:** ~100 mm (vs 85 mm standard)
- **Extra cost:** ~$2–4 per panel area (clay is typically free — excavated during foundation work)
- **STC:** ~45–50
- **Production change:** None to the panel itself. Clay layer is applied on-site after installation.

Best for: Budget-conscious upgrade. Uses free local material. No change to panel production. Adds humidity regulation as a bonus.

Option 3: Double Panel with Fiber Cavity (best acoustic performance, +15–20 STC)

Two standard panels with a 50–100 mm cavity between them, filled with loose organic fiber. This is the high-performance option for genuinely noisy environments.

Why This Works

Three acoustic principles combine:

1. **Double mass** — two mortar walls instead of one. The mass law says doubling mass adds ~6 dB of sound reduction.
2. **Decoupling** — the air cavity breaks the vibration path. Sound transmits efficiently through solid materials but poorly across an air gap. The two panels vibrate independently.
3. **Absorption** — loose fiber inside the cavity absorbs sound energy that would otherwise bounce between the two panels (standing wave resonance).

Cavity Fill Materials

Use whatever is locally available and cheap. The fill should be **loose, not compacted** — air pockets within the fiber are what absorb sound.

Material	Acoustic quality	Cost	Regional availability
Rice husk (*cascarilla de arroz*)	Excellent	Free–$2/bag	Rice-growing regions (Asia, Latin America)
Coconut coir (*fibra de coco*)	Excellent	$3–5/bag	Tropical coastal regions
Dried star grass (*pasto estrella*)	Good	Free	Tropical Americas (invasive, removed from fields)
Wood shavings (*viruta*)	Good	Free–$1/bag	Carpentry workshops

Material	Acoustic quality	Cost	Regional availability
Loose volcanic gravel (*gravilla volcánica*)	Good (mass-based)	$2–3/bag	Volcanic regions (Andes, Central America, SE Asia)
Dried banana leaf fiber	Good	Free	Banana-growing regions

Rice husk is the best all-round option where available: lightweight, doesn't compact over time, naturally fire-resistant (high silica content), and extremely cheap or free from rice mills.

Construction

1. Install the exterior panel as usual (bolted to building frame or ring beam)
2. Leave a 50–100 mm gap (set by spacers or a wider column/mounting bracket)
3. Fill the cavity loosely with fiber material — do not compact
4. Install the interior panel, closing the cavity
5. Both panels bolt to the same structural frame

The cavity width depends on the available space and desired performance:

Cavity width	STC improvement	Total wall thickness
50 mm	+12–15 STC	~220 mm
75 mm	+15–18 STC	~245 mm
100 mm	+18–20 STC	~270 mm

Specifications

- **Total wall thickness:** 220–270 mm (vs 85 mm standard)
- **Extra cost:** ~$65–70 per m² (one additional panel) + $2–5 for cavity fill = ~$70–75 extra per m²
- **STC:** 55–62
- **Panel production change:** None — uses two standard panels

Best for: Busy street frontage, highway-adjacent buildings, bedrooms facing noisy roads, music rooms, workshops adjacent to living spaces.

Combined Options

The options stack. For maximum acoustic performance on a critical wall:

Combination	STC	Extra cost/m²	Use case
Standard panel	40–45	Baseline	Quiet side
Dense mortar only	43–48	+$1–2	Mild noise, cheapest upgrade
Dense mortar + clay layer	48–52	+$3–6	Moderate street noise, single wall
Double panel + fiber cavity	55–62	+$70–75	Busy road
Double panel (dense mortar) + fiber cavity + clay interior	58–65	+$75–80	Highway, industrial, nightclub next door

Typical Building Example

A single-story house on a busy street corner, with garden on two sides:

Wall	Faces	Treatment	STC	Extra cost
North wall (10 m²)	Busy road	Double panel + rice husk	55–60	~$700–750
East wall (6 m²)	Side street	Dense mortar + clay layer	48–52	~$20–36
South wall (10 m²)	Garden	Standard panel	42–45	$0
West wall (6 m²)	Neighbor's garden	Standard panel	42–45	$0
Total extra cost				**~$720–790**

For ~$750, the noisiest wall gets professional-grade sound insulation while the quiet walls stay standard. The acoustic upgrade adds less than 10% to the total house cost from the cost estimate.

Material Notes

Fire Safety of Cavity Fill

- **Rice husk:** High silica content, naturally fire-resistant. Does not sustain flame. Smolders but self-extinguishes.
- **Coconut coir:** Slow to ignite, self-extinguishing when loose.
- **Dried grass/straw:** More flammable. Treat with borate solution (same as bamboo treatment) for fire resistance if using in the cavity. The cavity is sealed between two mortar panels, so ignition risk is very low regardless.
- **Volcanic gravel:** Non-combustible.

Moisture in the Cavity

The cavity must stay dry. Both panels are mortar-sealed and lime-plastered — moisture should not enter. However:

- Do not use the double-panel system below grade or in flood-prone areas
- Ensure the top of the cavity is sealed (ring beam or mortar cap)
- In extremely humid climates, leave small ventilation gaps at the top and bottom of the cavity (covered with mesh to prevent insect entry) to allow air circulation

Retrofit

The acoustic upgrade can be **added later** to an existing standard-panel wall:

1. Build a second panel wall 50–100 mm inside the existing wall
2. Fill the cavity with fiber
3. The interior room shrinks by the cavity width + panel thickness (~135–185 mm per upgraded wall)

This makes it possible to build standard panels now and upgrade specific walls later if noise becomes a problem — for example, if a quiet road becomes busy after development.